

Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank

A classification of the entire ring, its ideal theory,
and its behaviour under enlargement of the value group

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21 September 2026

Abstract

Let Γ be a nonzero set-sized divisible ordered abelian group and $K_\Gamma = \mathbb{C}((t^\Gamma))$ the full Hahn field over the trivially valued complex numbers. We study ordinary power series $\sum a_n Z^n$ whose evaluation families are strongly Hahn-summable at every point of this one fixed field. An exact coefficient criterion holds: this happens precisely when $v(a_n)/n$ tends cofinally to $+\infty$ in Γ . Consequently nonpolynomial entire series exist exactly when $\text{cf}(\Gamma) = \aleph_0$. Support-controlled preparation, proved by well-founded recursion over a well-ordered monoid rather than by any contraction mapping, yields exact root counts on every valuation ball, shell and residue direction, a genus-zero canonical product for every ball-finite divisor with no exponential correction factors, and complete factorization by zeros at arbitrary rank.

Two sharp classifications follow. The first is ideal-theoretic. The entire ring $\mathcal{E}_\Gamma$ is always a GCD domain with units $K_\Gamma^\times$; it is the polynomial ring, hence a PID, when $\text{cf}(\Gamma) > \aleph_0$; it is a non-Noetherian Bézout domain with unrestricted Hermite interpolation when Γ has an order unit; and otherwise it is a non-Noetherian GCD domain that is *not* Bézout, witnessed by two explicit entire functions with disjoint simple zero sets generating a proper ideal. A separate order-unit criterion decides invertibility in the valuation-restricted algebra $\mathcal{T}_\Gamma$, and shows that the positive case of the trichotomy is sharp rather than merely sufficient. The second classification is analytic. For every ordered-group extension $\Gamma \subseteq \Delta$, all nonpolynomial entire series over K_Γ share one exact strong evaluation domain in K_Δ : the valuation ring of v_Δ coarsened by the convex hull of Γ , whose residue field is $\mathbb{C}((t^H))$ rather than $\mathbb{C}$. Entireness survives exactly for cofinal extensions, no new zeros appear anywhere in the surviving domain, and on a noncofinal extension no alternative entire series agrees with the old one even on all ordinary complex constants. The scale boundary is intrinsic: the value group, the coefficient slopes and the negated zero valuations generate the same convex subgroup.

This report is the union of two independently written manuscripts that proved the same engine. The arbitrary-rank classifications, the explicit non-Bézout construction and the universal extension domain are proposed original contributions, not a claimed solution of any named published conjecture. Classical Hahn-field inputs and rank-one antecedents are identified and credited. Complete proofs are supplied; they have not been independently refereed or verified in a proof assistant.

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1 The question, the answer, and its scope

1.1 Two questions that turn out to have one engine

Over an algebraically closed field, finite polynomials with disjoint zero sets generate the unit ideal. In familiar one-variable analytic theories, interpolation extends this fact to entire functions, and it is tempting to infer the same conclusion as soon as an infinite product theorem is available. The inference is false in the higher-rank setting studied here. We obtain a ring in which every admissible zero divisor is realized, every function factors completely according to its zeros, and greatest common divisors exist, yet two functions without a common zero need not satisfy a Bézout identity.

The second question is what the word *entire* names. A formal series with coefficients in one Hahn field may be evaluable at every point of that field and lose even strong summability at a single point of a larger one. Both questions are decided by the same object: the ordered hierarchy of scales above the coefficient slopes of the series. The first is decided by whether those scales have a greatest Archimedean class; the second by whether they remain cofinal after the enlargement.

The obstruction behind the first question is genuinely infinite. At increasingly large arguments, any one entire function is restricted by a common lower bound on the valuations of its coefficients. We place a second sequence of zeros so close to the first that the reciprocal values required by a Bézout identity escape those bounds. Every finite interpolation problem remains solvable by a polynomial. It is the simultaneous infinite problem that fails.

1.2 The analytic category

Throughout, Γ is a *set-sized divisible ordered abelian group*, written additively. Except where the degenerate case $\Gamma = 0$ is explicitly discussed, assume $\Gamma \neq 0$. Put

$$K_\Gamma = \mathbb{C}((t^\Gamma)), \quad v\left(\sum_{\eta} c_\eta t^\eta\right) = \min\{\eta : c_\eta \neq 0\}.$$

A Hahn series has well-ordered support. We put $v(0) = +\infty$. This is the *full* Hahn field: all well-ordered supports are admitted, not a support-restricted subfield. The complex coefficient field carries the *trivial* valuation, so every nonzero complex coefficient has valuation zero and an infinite ordinary complex sum at one exponent is not an allowed Hahn coefficient operation. The valuation ring and its maximal ideal are

$$\mathcal{O}_\Gamma = \{x \in K_\Gamma : v(x) \geq 0\}, \quad \mathfrak{m}_\Gamma = \{x \in K_\Gamma : v(x) > 0\},$$

with residue field exactly $\mathbb{C}$.

Definition 1.1. The ring of *strongly entire Hahn functions* is

$$\mathcal{E}_\Gamma = \left\{ F = \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]] : (a_n z^n)_{n \geq 0} \text{ is strongly Hahn-summable for every } z \in K_\Gamma \right\}.$$

Here strong summability means that the union of the term supports is well ordered and each exponent belongs to only finitely many term supports. That this set is a ring is proved below.

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. The variable Z is an ordinary single indeterminate; its exponents are nonnegative *integers*. The Hahn exponents occur only in the coefficients, and the two infinities must

not be confused: a finite index n in a power series is not a Hahn exponent, and integer multiples of a positive Hahn exponent need not be cofinal. Nothing is asserted about arbitrary generalized series in Z with ordinal support order types, about arbitrary local functions for the fine surreal topology, or about the Berarducci–Mantova derivation. All derivatives in this report hold K_Γ constant.

Convention 1.2 (Ring discipline). Every numbered statement below is asserted over one named coefficient ring, and no statement is transferred to another. The rings used here are

$$K_\Gamma[Z] \subseteq \mathcal{E}_\Gamma \subseteq \mathcal{T}_\Gamma \subseteq \mathcal{A}_\Gamma \subseteq K_\Gamma[[Z]],$$

defined in [sections 2](#) and [4](#), all of which have *Hahn series over $\mathbb{C}$* as coefficients. In particular no theorem here is a theorem about $\mathcal{O}(U)((t^\Gamma))$ or $\mathbb{C}\{z\}((t^\Gamma))$, whose coefficients are ordinary holomorphic functions on a common complex domain or convergent germs. That distinction is not a formality: the companion report *Analytic Geometry over Surcomplex Hahn Fields* adopts the same discipline by name for its own four coefficient rings, and records there an explicit element of the radius-free algebra lying in no common-domain algebra. We invoke that convention here so that no reader transports a statement of this report into those categories. See [section 13](#).

1.3 The two classifications

A subset of Γ is cofinal if every element of Γ is at most some member of that subset; $\text{cf}(\Gamma)$ is the least cardinality of a cofinal subset, and it is at least $\aleph_0$ for $\Gamma \neq 0$, since an ordered group has no greatest element.

Definition 1.3 (Order unit). An element $\delta > 0$ of Γ is an *order unit* if for every $\gamma \in \Gamma$ some positive integer m satisfies $m\delta > \gamma$; that is, $\{m\delta : m \geq 1\}$ is cofinal in Γ .

Remark 1.4 (One name for one invariant). This report uses exactly two order-theoretic invariants of Γ , and they must not be conflated.

- *Cofinality* always means $\text{cf}(\Gamma)$, the least cardinality of a cofinal subset of the underlying ordered set of the value group. It is never the cofinality of an ordinal index, of a diagram, or of the image of some displacement or spectral map; other reports in this collection use the word for such other invariants, and nothing here transfers to them.
- *Order unit* is the condition of [definition 1.3](#). The two source manuscripts merged here named it twice: one called δ an order unit, the other called it a *cofinal-cyclic element*, meaning exactly that the cyclic subgroup it generates is cofinal. These are the same condition, stated twice; we keep the single name *order unit* and record the synonym here once. For a fixed $\delta > 0$, it is equivalent to each of the following: δ represents the greatest positive Archimedean class of Γ ; $m\delta \rightarrow +\infty$; the monomial t^δ is topologically nilpotent in the intrinsic valuation topology of K_Γ . Thus an order unit exists exactly when the positive Archimedean classes have a greatest member; a positive element in a smaller class is not an order unit.

An order unit forces $\text{cf}(\Gamma) = \aleph_0$, but not conversely: the group Γ_∞ of [section 12](#) has countable cofinality and no order unit. That gap is exactly the third case of the trichotomy below, and also the difference between the two existence questions of [theorems 1.5](#) and [4.1](#).

Theorem 1.5 (Cofinality–rank trichotomy). *Let $\Gamma \neq 0$ be a divisible ordered abelian group. Then $\mathcal{E}_\Gamma$ is a GCD domain, its units are exactly $K_\Gamma^\times$, and exactly one of the following cases occurs.*

- (i) *If $\text{cf}(\Gamma) > \aleph_0$, then $\mathcal{E}_\Gamma = K_\Gamma[Z]$, a principal ideal domain.*
- (ii) *If Γ has an order unit, then $\mathcal{E}_\Gamma$ contains nonpolynomial functions and is a non-Noetherian Bézout domain. Arbitrary finite-order jets on every radially finite set can be interpolated by an entire function.*
- (iii) *If $\text{cf}(\Gamma) = \aleph_0$ and Γ has no order unit, then $\mathcal{E}_\Gamma$ contains nonpolynomial functions and is a non-Noetherian GCD domain that is not Bézout. More strongly, there are $f, g \in \mathcal{E}_\Gamma$ with disjoint simple zero sets and $(f, g) \neq \mathcal{E}_\Gamma$.*

For $\Gamma = 0$, the strong-summation convention gives $\mathcal{E}_0 = \mathbb{C}[Z]$.

A GCD domain is an integral domain in which every pair has a greatest common divisor, defined up to a unit. A Bézout domain is one in which every finitely generated ideal is principal. Case (iii) therefore separates two properties that coincide for finite polynomial algebra. This is not a distinction about algebraic closedness: every K_Γ here is algebraically closed.

Theorem 1.6 (Universal scalar-extension package). *Let $\Gamma \subseteq \Delta$ be nonzero divisible ordered abelian groups and let $H = \text{conv}_\Delta(\Gamma)$ be the convex hull of Γ in Δ . Write $\mathcal{D}_{\Gamma, \Delta}$ for the valuation ring of v_Δ coarsened by H . Then for every nonpolynomial $f \in \mathcal{E}_\Gamma$:*

- (a) *the family $(a_n z^n)_n$ is strongly summable in K_Δ exactly for $z \in \mathcal{D}_{\Gamma, \Delta}$, so the domain does not depend on f , and $f(\mathcal{D}_{\Gamma, \Delta}) \subseteq \mathcal{D}_{\Gamma, \Delta}$;*
- (b) *$\mathcal{D}_{\Gamma, \Delta} = K_\Delta$, equivalently $f \in \mathcal{E}_\Delta$, exactly when Γ is cofinal in Δ ;*
- (c) *every zero of the extended function on $\mathcal{D}_{\Gamma, \Delta}$ already lies in K_Γ with unchanged multiplicity, even when f has infinitely many zeros;*
- (d) *if Γ is not cofinal in Δ , there is no $g \in \mathcal{E}_\Delta$ agreeing with f on $\mathbb{C}$, let alone on K_Γ ;*
- (e) *Γ , the slope set $\{v(a_n)/n : n \geq 1, a_n \neq 0\}$ and the negated zero valuations $\{-v(r) : r \neq 0, f(r) = 0\}$ generate the same convex subgroup of any larger ordered group, so $\mathcal{D}_{\Gamma, \Delta}$ is recoverable from the coefficients alone or from the zeros alone.*

The residue field of $\mathcal{D}_{\Gamma, \Delta}$ is canonically $\mathbb{C}((t^H))$, not $\mathbb{C}$.

Remark 1.7 (The two extension statements are one statement). Part (b) is often phrased negatively — in the noncofinal case the series is simply *not entire over K_Δ* , and indeed $\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = K_\Gamma[Z]$ there (theorem 10.1) — while parts (a) and (c) say that the very same series remains strongly summable on all of $\mathcal{D}_{\Gamma, \Delta}$ and keeps every one of its zeros there. Read side by side these look like a disagreement about whether the function survives an enlargement. They are not in conflict: they are the coarse and the exact form of one fact. “Not entire over K_Δ ” means precisely that the strong evaluation domain is a *proper* subring of K_Δ , and theorem 10.4 computes exactly which subring it is, namely $\mathcal{D}_{\Gamma, \Delta}$, on which the function is unchanged. The negative statement locates the boundary; the positive statement describes everything inside it. Throughout this report “entire over K ” is reserved for strong summability at *every* point of K , and any weaker survival is described by naming its domain.

The proof of [theorem 1.5](#) is distributed through the article. The coefficient criterion and cofinality dichotomy are in [section 3](#); the order-unit criterion for the restricted algebra, which is what makes case (ii) sharp rather than merely sufficient, is in [section 4](#); preparation and root counting are in [sections 5 and 6](#); factorization and GCDs are in [section 7](#); the negative and positive Bézout results are in [sections 8 and 9](#). [Section 11](#) assembles them. [Theorem 1.6](#) is proved in [section 10](#).

1.4 Relationship to the repository and the literature

The repository supplied with the question already contains a rank-one report over $\mathbb{C}((t^{\mathbb{R}}))$, including preparation, zero counting, zero-free rigidity and a partial-theta canonical product [\[14\]](#). Neither source manuscript claims to originate those points, and neither do we; the present report is their arbitrary-rank generalization, and the rank-one report itself proves the obstruction that makes rank one non-generic. The repository also contains a global-divisor report in a different ring, $\mathcal{O}(U)((t^{\Gamma}))$, with ordinary holomorphic coefficient functions on a common complex domain [\[15\]](#); its global support obstructions belong to a different analytic category and our argument is not a restatement of its Picard-group or moving-divisor results. [Section 13](#) states the cross-references in full.

Classical non-Archimedean factorization is an essential antecedent. Lazard’s work is a foundational reference for one-variable zeros [\[5\]](#); Cherry records the one-variable product factorization and develops GCD and factorization results under a complete real-valued non-Archimedean absolute value [\[6, 7\]](#), and himself calls those factorization results well known. Hahn-field algebraic closedness is standard, used here in the form of Poonen’s Corollary 4 [\[4\]](#); the support calculus is Neumann’s [\[3\]](#); support-based operator methods are developed extensively by van der Hoeven [\[9\]](#). The distinction between positive valuation and topological nilpotence in rank two is established, and we follow Conrad’s discussion of it [\[8\]](#). We claim none of these facts as new.

Novelty statement. The proposed contributions are the arbitrary-rank coefficient, unit and workspace criteria, the explicit non-Bézout construction, the universal scalar-extension domain with zero conservation, and their combination into [theorems 1.5 and 1.6](#). The support-controlled preparation and product arguments provide a self-contained route to those results; their rank-one specializations are not new. The literature checks performed for the two source manuscripts did not locate the stated arbitrary-rank classifications or the explicit obstruction in the primary sources consulted. This is a bounded literature statement, not a certification of priority. The repository review was targeted at the catalogue, relevant READMEs and stated scope, not a line-by-line audit of every manuscript. No named published open problem is claimed solved. [Section 14](#) and [appendix D](#) state the limits in full; every limitation recorded by either source manuscript is preserved there.

2 Hahn supports, the surreal interpretation, and the algebra chain

2.1 Foundational setting

The proofs can be carried out in ZFC using only the set-sized field K_{Γ} . To interpret them inside the surcomplex numbers, take Γ to be a divisible ordered additive subgroup of $\mathbf{No}$ and send

$$\sum_{\gamma} c_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} c_{\gamma} \omega^{-\gamma}. \quad (1)$$

The convention $t^{\gamma} := \omega^{-\gamma}$ is Conway’s monomial map; it is explicitly *not* a surreal exponential. The real and imaginary parts of the image are Conway normal forms, and the standard normal-form

arithmetic makes (1) an embedding of fields: the real coefficient subfield $\mathbb{R}((t^\Gamma))$ lies in $\mathbf{No}$, and its coefficientwise complexification lies in $\mathbf{No}[i]$ [1, 2]. Conjugation acts on coefficients only. Our explicit examples use displayed subgroups of $\mathbf{No}$, so they do not require an unspecified universal embedding theorem. An element $x \in \mathcal{O}_\Gamma$ has a unique expression $x = c + \varepsilon$ with $c \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}_\Gamma$; the residue map $\text{res}(x) = c$ is defined on $\mathcal{O}_\Gamma$, not on all of K_Γ .

2.2 The support facts actually used

We use the following classical Hahn–Neumann support facts [3][4, Section 3][9]. They are stated explicitly because every infinite construction below depends on them, and they are imported, not proved here.

Lemma 2.1 (Hahn–Neumann support calculus). *Let S, T be well-ordered subsets of an ordered abelian group.*

- (a) *$S \cup T$ and $S + T$ are well ordered. For every η , there are only finitely many pairs $(s, t) \in S \times T$ with $s + t = \eta$.*
- (b) *If $S \subseteq \Gamma_{>0}$, its additive monoid $S^* = \{s_1 + \cdots + s_r : r \geq 0, s_i \in S\}$ is well ordered. For each η , there are only finitely many finite ordered words in S whose sum is η .*
- (c) *Consequently, if u is a Hahn series over a commutative coefficient ring and $v(u) > 0$, the geometric series $\sum_{r \geq 0} u^r$ is strongly summable and is the inverse of $1 - u$.*

Part (b) is a support theorem, not an assertion that $nv(u)$ tends cofinally to infinity. The latter can fail. In $\mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $e_2 > ne_1$ for every n , the Hahn identity $\sum_{n \geq 0} t^{ne_1} = (1 - t^{e_1})^{-1}$ is valid but its partial sums do not converge in the intrinsic valuation topology: no tail has valuation greater than e_2 . [Example 4.3](#) turns this observation into a theorem about units.

Lemma 2.2 (Cofinal tails). *Suppose $u_n \in K_\Gamma$ and $v(u_n) \rightarrow +\infty$ cofinally in Γ ; that is, for every $B \in \Gamma$, eventually $v(u_n) > B$. Then (u_n) is strongly summable. Its sum has valuation at least $\min_n v(u_n)$ when at least one term is nonzero.*

Proof. For a fixed B , only finitely many nonzero term supports meet $(-\infty, B]$. A descending sequence in their union, beginning at B , would therefore lie in a finite union of well-ordered supports, which is impossible. Likewise, any exponent occurs in only finitely many term supports. The valuation estimate follows from coefficientwise addition. A nonempty set of term valuations has a minimum because it is a subset of the well-ordered support union. $\square$

We use the usual ZFC equivalence between well-ordering of a linear order and the absence of an infinite strictly descending sequence. All supports and index sets are sets; no proper-class support or proper-class summation is involved.

2.3 Three meanings of an infinite expression

It is essential to keep separate

- (1) a strong Hahn sum, which requires only a well-ordered support union and pointwise finiteness;
- (2) convergence in the intrinsic valuation topology of the fixed field K_Γ , which requires cofinal progress through every threshold of Γ ;

(3) convergence in the topology inherited from all surreal scales.

These need not agree. [Definition 1.1](#) starts with (1). The *all-point* condition will force (2) for evaluation series ([theorem 3.1](#)), even though (1) alone does not, and on a single disk the two genuinely differ. We make no assertion of convergence in (3), and no density of $\mathbb{C}$ in any fine surreal topology is asserted or used. The repository's foundational reports emphasize this distinction [\[13, 14\]](#).

2.4 The chain of coefficient algebras

Four algebras of series in Z over K_Γ are kept distinct throughout. Define the *polynomial-coefficient Hahn algebra*

$$\mathcal{A}_\Gamma = \mathbb{C}[X]((t^\Gamma)), \quad (2)$$

whose elements are $A = \sum_{\eta \in S} P_\eta(X)t^\eta$ with S well ordered and every P_η a finite polynomial, with no common bound on $\deg P_\eta$; and the *valuation-restricted algebra*

$$\mathcal{T}_\Gamma = \left\{ \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]] : v(a_n) \rightarrow +\infty \text{ cofinally in } \Gamma \right\}. \quad (3)$$

Taking the coefficient of each X^n gives an injective ring map $\mathcal{A}_\Gamma \hookrightarrow K_\Gamma[[X]]$, and under it $\mathcal{A}_\Gamma$ is exactly the set of series whose coefficient family $(a_n)_n$ is itself strongly summable: finite degree at each Hahn exponent is precisely the pointwise-finiteness half of strong summability. With X and Z identified under that map,

$$K_\Gamma[Z] \subseteq \mathcal{E}_\Gamma \subseteq \mathcal{T}_\Gamma \subseteq \mathcal{A}_\Gamma \subseteq K_\Gamma[[Z]]. \quad (4)$$

The first two inclusions need not be strict; the last two record genuinely different summation requirements. The two source manuscripts named the ring (2) differently — one as the auxiliary algebra $\mathcal{A}_\Gamma$ in the variable X , the other as the strong disk algebra $\mathcal{H}_\Gamma$ in the variable Z — but it is one ring, and we use the single symbol $\mathcal{A}_\Gamma$. That the inclusions in (4) hold, and that $\mathcal{T}_\Gamma$ and $\mathcal{A}_\Gamma$ are rings, follows from [lemma 2.1](#) together with [proposition 3.5](#) below.

For $A \neq 0$ in $\mathcal{A}_\Gamma$, set $w(A) = \min\{\eta : P_\eta \neq 0\}$; the polynomial $P_{w(A)}$ is its *initial polynomial*. Since $\mathbb{C}[X]$ is a domain,

$$w(AB) = w(A) + w(B), \quad (5)$$

and initial polynomials multiply. Equivalently, writing $\text{in}_0(A) = \overline{t^{-w(A)}A} \in \mathbb{C}[X] \setminus \{0\}$ for the reduction, $\text{in}_0(AB) = \text{in}_0(A)\text{in}_0(B)$. This Gauss multiplicativity is the engine of [section 4](#).

3 An exact coefficient criterion and the cofinality dichotomy

For $n \geq 1$, division of $v(a_n)$ by n is legitimate because Γ is divisible and torsion-free. We interpret $(+\infty)/n = +\infty$, so that zero coefficients impose no condition.

Theorem 3.1 (All-point strong summability). *Let $\Gamma \neq 0$ and $F = \sum_{n \geq 0} a_n Z^n \in K_\Gamma[[Z]]$. The following conditions are equivalent.*

- (1) $F \in \mathcal{E}_\Gamma$.

- (2) For every $\gamma \in \Gamma$, the family $(a_n t^{n\gamma})_{n \geq 0}$ is strongly summable; equivalently $F(t^\gamma X) \in \mathcal{A}_\Gamma$.
- (3) For every $h \in \Gamma$, eventually $v(a_n) > nh$.
- (4) For every $\gamma, B \in \Gamma$, eventually $v(a_n) + n\gamma > B$.

In particular, strong entire evaluation is equivalent to the cofinal condition $v(a_n)/n \rightarrow +\infty$, and it is enough to test the monomial points $z = t^\gamma$.

Proof. The implication (1) $\Rightarrow$ (2) is immediate by testing $z = t^\gamma$; the reformulation of (2) inside $\mathcal{A}_\Gamma$ holds because the common Hahn support is the union of the supports of the scaled coefficients, and the coefficient at a fixed Hahn exponent is a polynomial in X exactly when that exponent occurs at only finitely many indices n . The zero series satisfies all four conditions, so assume $F \neq 0$.

Suppose (2) holds. Applying it at $\gamma = 0$ shows that the union of the supports of the a_n is well ordered. Thus

$$m = \min\{v(a_n) : a_n \neq 0\} \in \Gamma$$

exists. Suppose (3) fails. There is $h \in \Gamma$ for which infinitely many positive integers n satisfy $a_n \neq 0$ and $v(a_n) \leq nh$. Choose $q > 0$ with $q > |h|$ and $q > |m|$, where $|\cdot|$ is the ordered group absolute value. By (2), the set of leading exponents

$$\{v(a_n) - 2nq : a_n \neq 0\}$$

is well ordered. Its minimum is attained at some *finite* index k ; write that minimum as μ . Then

$$\mu = v(a_k) - 2kq \geq m - 2kq > -(2k+1)q.$$

Choose a violating index $n > 2k+1$. It gives

$$v(a_n) - 2nq \leq nh - 2nq < -nq < -(2k+1)q < \mu,$$

a contradiction. This proves (2) $\Rightarrow$ (3). Notice that the argument used a minimum attained at a finite index; it did not assume that the multiples of q are cofinal in Γ . See [appendix A](#).

Suppose (3) holds and fix γ, B . Choose $r > 0$ and set $h = -\gamma + |B| + r$. Eventually

$$v(a_n) + n\gamma > n(|B| + r) > B$$

for $n \geq 1$. Thus (4) holds.

Finally, fix $z \neq 0$ and put $\gamma = v(z)$. Condition (4) implies $v(a_n z^n) = v(a_n) + n\gamma \rightarrow +\infty$. Apply [lemma 2.2](#). At $z = 0$ there is only the constant term. This proves (4) $\Rightarrow$ (1). $\square$

Remark 3.2. Condition (3) is strictly stronger than $v(a_n) \rightarrow +\infty$, that is, than membership in $\mathcal{T}_\Gamma$. Over $\Gamma = \mathbb{Q}$ the coefficients $a_n = t^n$ tend to zero in valuation, but $\sum t^n Z^n$ is not entire: at $Z = t^{-1}$ its terms all have exponent zero. Conversely $a_n = t^{n^2}$ satisfies (3) over $\mathbb{Q}$ and over $\mathbb{R}$, but not over a group containing an element larger than every real number. That last example is exactly the one recorded in the foundations report of this collection; see [section 13](#).

Corollary 3.3 (Cofinality dichotomy). *For $\Gamma \neq 0$, the following are equivalent: $\mathcal{E}_\Gamma \neq K_\Gamma[Z]$; $\mathcal{E}_\Gamma$ contains a nonpolynomial series; Γ has a countable cofinal subset; $\text{cf}(\Gamma) = \aleph_0$. If $\text{cf}(\Gamma) > \aleph_0$, every strongly entire ordinary power series is a polynomial.*

Proof. If infinitely many a_n are nonzero, their valuations divided by their positive indices give, by [theorem 3.1](#), a countable cofinal subset of Γ . Conversely, choose a positive increasing cofinal sequence $(q_n)_{n \geq 1}$. Then

$$1 + \sum_{n \geq 1} t^{nq_n} Z^n$$

satisfies the coefficient criterion and is nonpolynomial. A nonzero ordered group has no finite cofinal subset, because it has no greatest element. $\square$

Example 3.4 (Two different sources of polynomial rigidity). If $\Gamma = 0$, every nonzero coefficient has support $\{0\}$, so strong summability at 1 forces a polynomial. If Γ has uncountable cofinality, there are many infinitesimal coefficients, but countably many coefficient slopes cannot reach all of its scales. The conclusion is again polynomiality, for a different reason. The same countability argument shows that in the uncountable-cofinality case even the larger algebra $\mathcal{T}_\Gamma$ collapses to $K_\Gamma[Z]$; see [\(43\)](#).

Proposition 3.5 (Entire operations). $\mathcal{E}_\Gamma$ is a K_Γ -subalgebra of $K_\Gamma[[Z]]$, and so are $\mathcal{T}_\Gamma$ and $\mathcal{A}_\Gamma$. Evaluation at each $z \in K_\Gamma$ is a homomorphism on $\mathcal{E}_\Gamma$. Translation $F(Z) \mapsto F(Z+b)$, differentiation, and the primitive with zero constant coefficient preserve $\mathcal{E}_\Gamma$. The translation coefficients are

$$[Z^k]F(Z+b) = \sum_{n \geq k} \binom{n}{k} a_n b^{n-k} = \frac{F^{(k)}(b)}{k!}. \quad (6)$$

Entire composition $F \circ G$ is also defined and belongs to $\mathcal{E}_\Gamma$. Moreover, at each fixed scale $s \in \Gamma$ the partial sums of $F \in \mathcal{E}_\Gamma$ converge to F in the weighted Gauss valuation m_s of [\(13\)](#).

Proof. The ring assertion for $\mathcal{E}_\Gamma$ follows from [theorem 3.1\(2\)](#): addition and multiplication preserve every rescaled copy of $\mathcal{A}_\Gamma$. Explicitly, at a fixed s write $A_i = v(a_i) + is$ and $B_j = v(b_j) + js$ for two members; both tend cofinally to infinity and have lower bounds A, B . Given β , choose N with $A_i > \beta - B$ and $B_i > \beta - A$ for $i > N$; if $n > 2N$ then each pair $i + j = n$ has $i > N$ or $j > N$, so the convolution coefficient has weighted valuation greater than β . Taking $s = 0$ gives the ring assertion for $\mathcal{T}_\Gamma$; the same tail estimate gives the stated weighted convergence of partial sums. That $\mathcal{A}_\Gamma$ is a ring is [lemma 2.1\(a\)](#).

Evaluation can be performed by choosing $\gamma = v(z)$, writing $z = t^\gamma x$ with $x \in \mathcal{O}_\Gamma$, and applying [lemma 5.1](#).

For translation and a fixed testing scale γ , expand $F(b + t^\gamma X)$ into its finite binomial blocks. Put $\beta = \min\{v(b), \gamma\}$, with $v(0) = +\infty$. Every coefficient in the n th block has valuation at least $v(a_n) + n\beta$, which tends cofinally to infinity, and each block has finitely many X -coefficients. The support-union and local-finiteness proof of [lemma 2.2](#) therefore applies to the blocks with polynomial coefficients. This gives an element of $\mathcal{A}_\Gamma$ at every scale and proves [\(6\)](#). Regrouping the same jointly summable family proves the evaluation identity and the inverse translation by $-b$.

For differentiation, $v(na_n) = v(a_n)$ for positive n , since n is a nonzero complex coefficient of valuation zero. Shifting the index by one preserves the criterion in [theorem 3.1](#): for a fixed h , choose $q > 0$ with $q > |h|$; the bound $v(a_n) > n(h+q)$ for large n implies the required bounds with $n-1$ or $n+1$ in place of n . No hidden Archimedean estimate is used. Division of a_n by $n+1$ also preserves valuation, so termwise integration preserves the criterion as well.

For composition, fix γ and let $G_\gamma = G(t^\gamma X) \in \mathcal{A}_\Gamma$. If $G_\gamma \neq 0$, put $\mu = w(G_\gamma)$. Then $w(a_n G_\gamma^n) = v(a_n) + n\mu \rightarrow +\infty$, so the sum is strongly summable in $\mathcal{A}_\Gamma$. Its coefficients define the same formal composite at every scale, since rescaling commutes with each coefficient's strong sum.

The case $G = 0$ is constant. Evaluation and associativity follow from the same support-controlled regrouping. $\square$

4 Restricted series: the sharp order-unit criterion

The trichotomy of [theorem 1.5](#) splits its two nonpolynomial cases by the presence of an order unit. This section proves that the order unit is not an artefact of the proof technique of [section 9](#): it is exactly the condition for a nonconstant element of the restricted algebra $\mathcal{T}_\Gamma$ of (3) to be invertible there. That is what makes case (ii) of the trichotomy sharp rather than merely sufficient, and it is a statement about $\mathcal{T}_\Gamma$, not about $\mathcal{E}_\Gamma$; the ring discipline of [convention 1.2](#) applies.

Theorem 4.1 (Restricted-unit criterion). *Let $f = \sum a_n Z^n \in \mathcal{T}_\Gamma$ be nonconstant. Then f is a unit of $\mathcal{T}_\Gamma$ if and only if $a_0 \neq 0$ and*

$$\delta(f) = \min_{n \geq 1, a_n \neq 0} v(a_n/a_0) \quad (7)$$

is a positive order unit of Γ . Nonzero constant series are always units, and the minimum in (7) exists whenever $a_0 \neq 0$.

Proof. The minimum exists because the coefficient valuations tend cofinally to infinity: below the valuation of any chosen nonzero coefficient there are only finitely many nonzero coefficients.

Suppose $fg = 1$ with $g \in \mathcal{T}_\Gamma$. The constant coefficient forces $a_0 \neq 0$. Divide f by a_0 and multiply g by a_0 , so that both constant coefficients are 1. Then $w(f), w(g) \leq 0$, while (5) gives their sum as zero. Hence both are zero. Their reductions in $\mathbb{C}[X]$ multiply to 1, so both reductions are constant. Every nonconstant coefficient of f and of g consequently has positive valuation; in particular $\delta = \delta(f) > 0$.

Assume for contradiction that δ is not an order unit. The convex subgroup it generates,

$$H_\delta = \{\gamma \in \Gamma : |\gamma| \leq m\delta \text{ for some } m \in \mathbb{N}\}, \quad (8)$$

is then proper. Coarsen the valuation to the ordered quotient Γ/H_δ . Every coefficient of f and g is integral for this coarsening. Their coarsened reductions are *polynomials* over the coarsened residue field: choose $\beta > H_\delta$, meaning $\beta > h$ for every $h \in H_\delta$; since coefficient valuations tend cofinally to infinity in Γ , only finitely many coefficients can have value at most β , and all the remaining ones reduce to zero.

The reduced polynomial of f is nonconstant, because a nonconstant coefficient of valuation $\delta \in H_\delta$ has nonzero coarsened residue. On the other hand the reduced polynomials of f and g multiply to 1 in a polynomial ring over a field, and a nonconstant polynomial cannot be a unit there. This contradiction proves that δ is an order unit.

Conversely, suppose $a_0 \neq 0$ and $\delta(f)$ is a positive order unit. Write $f = a_0(1 + h)$, where h has zero constant coefficient and $w(h) = \delta$. The formal inverse is

$$f^{-1} = a_0^{-1} \sum_{m \geq 0} (-h)^m. \quad (9)$$

Its summands have Gauss valuation $m\delta \rightarrow +\infty$, so the sum exists strongly in $\mathcal{A}_\Gamma$. To prove that it belongs to $\mathcal{T}_\Gamma$, fix a valuation threshold β . All sufficiently large m contribute only coefficients of valuation greater than β ; the remaining finitely many powers of h belong to $\mathcal{T}_\Gamma$, so their coefficients also eventually exceed β . Thus the coefficients of (9) tend cofinally to infinity. The finite geometric identity, followed by the justified strong sum, proves $ff^{-1} = 1$. $\square$

The coarsening argument is what makes this a *necessity* theorem rather than the observation that one particular geometric-series construction fails. It shows that no cancellation in a hypothetical inverse can evade the order-unit obstruction.

Corollary 4.2 (Unit group of the restricted algebra). *The algebra $\mathcal{T}_\Gamma$ has a nonconstant unit if and only if Γ has an order unit; otherwise $\mathcal{T}_\Gamma^\times = K_\Gamma^\times$. The rule “the constant coefficient strictly dominates all the others” is a sufficient unit criterion for every restricted series if and only if Γ is Archimedean.*

Proof. An order unit δ makes $1 - t^\delta Z$ a nonconstant unit by the theorem, and the converse follows by applying the theorem to any nonconstant unit. An ordered group is Archimedean exactly when every positive element has cofinal positive integer multiples; apply that characterization to $1 - t^\delta Z$ for the last assertion. $\square$

Example 4.3 (Zero-free does not mean restricted-invertible). Let $\Gamma = \mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$ with $0 < \epsilon \ll \eta$, the η coordinate being compared first. The polynomial

$$f(Z) = 1 - t^\epsilon Z$$

has residue 1 and no zero in $\mathcal{O}_\Gamma$: its unique zero $t^{-\epsilon}$ has negative valuation. Yet its only formal inverse, $\sum_{n \geq 0} t^{n\epsilon} Z^n$, is not restricted, since $n\epsilon < \eta$ for every n . It does belong to $\mathcal{A}_\Gamma$ and can be strongly evaluated on the integral disk. By contrast $1 - t^\eta Z$ has a restricted inverse, because $n\eta$ is cofinal.

Remark 4.4 (Why a rank-one contraction proof may not be imported). [Example 4.3](#) is the reason the preparation theorem of [section 5](#) is proved by well-founded recursion over a well-ordered monoid and not by a convergent iteration. A rank-one Banach argument containing the phrase “the valuation increases by a positive amount at each step” is not valid here: the standard iteration may improve the error by ϵ repeatedly while remaining forever below η , and its purported convergence is then simply unproved. For a valid valuation-convergence proof the positive amount must be an order unit. The preparation proof below uses positive *supports* and finite-word control instead, and works with no order unit at all. The same care applies to [section 9](#), whose real-valued coarsening is constructed only after an order unit has been assumed.

Remark 4.5 (Two existence questions, genuinely different). Nonpolynomial entire series need only $\text{cf}(\Gamma) = \aleph_0$ ([corollary 3.3](#)); a nonconstant restricted unit needs an order unit ([corollary 4.2](#)). The group Γ_∞ of [section 12](#) has the first and not the second, so the two questions separate. In topological language: the valuation topology of K_Γ has a countable neighbourhood base at zero exactly when $\text{cf}(\Gamma) = \aleph_0$, whereas a nonzero topologically nilpotent element exists exactly when Γ has an order unit, since $v(x^n) = nv(x)$. Neither of these underlying topological facts is claimed as new; the higher-rank framework in which they sit is discussed by Conrad [\[8\]](#).

5 Support-controlled preparation

5.1 The algebra that retains support certificates

Recall $\mathcal{A}_\Gamma = \mathbb{C}[X]((t^\Gamma))$ from [\(2\)](#). The support facts in [lemma 2.1](#) justify multiplication in $\mathcal{A}_\Gamma$ and show that the coefficient map $\mathcal{A}_\Gamma \hookrightarrow K_\Gamma[[X]]$ respects it.

Lemma 5.1 (Units and integral substitution). *A nonzero element of $\mathcal{A}_\Gamma$ is a unit if and only if its initial polynomial is a nonzero constant. Every $A \in \mathcal{A}_\Gamma$ can be evaluated at any $x \in \mathcal{O}_\Gamma$, and this evaluation is a ring homomorphism; if $A \neq 0$ the value has valuation at least $w(A)$. For $x \in \mathcal{O}_\Gamma$, substitution $X \mapsto X + x$ is an automorphism of $\mathcal{A}_\Gamma$.*

Proof. If the initial polynomial is $c \neq 0$ at exponent η , write $A = ct^\eta(1 + u)$ with $w(u) > 0$ and invert $1 + u$ using lemma 2.1(c). Conversely, if $AB = 1$, the product of the two initial polynomials is 1, so both are nonzero constants.

For substitution, put $S = \text{supp}_t A$ and write $x = c + \varepsilon$ with $c \in \mathbb{C}$ and $v(\varepsilon) > 0$, treating $\varepsilon = 0$ separately. For each η , expand the finite polynomial $P_\eta(X + c + \varepsilon)$ in powers of ε . All supports lie in $S + (\text{supp } \varepsilon)^*$, which is well ordered. At a fixed exponent there are only finitely many decompositions from $S + (\text{supp } \varepsilon)^*$, finitely many words in $\text{supp } \varepsilon$ for each such decomposition, and finitely many terms from each P_η . Thus substitution is a legitimate element of $\mathcal{A}_\Gamma$. Finite polynomial identities and strong regrouping prove multiplicativity, and its inverse is substitution by $-x$. Evaluation follows by setting $X = 0$ after this substitution. If $A \neq 0$, every contribution has exponent at least $\min S = w(A)$, because $v(\varepsilon) > 0$; cancellation can only raise the valuation. This gives the claimed bound without any assertion of cofinal tail convergence. $\square$

5.2 Preparation without an Archimedean contraction

Theorem 5.2 (Support-controlled preparation). *Let $A \in \mathcal{A}_\Gamma$ satisfy $w(A) = 0$, and suppose its initial polynomial $P \in \mathbb{C}[X]$ is monic of degree d . There are unique W, U such that*

$$A = WU, \quad W = P + R, \quad R \in \mathfrak{m}_\Gamma[X], \quad \deg R < d, \quad U = 1 + Q, \quad w(Q) > 0.$$

Here W is a monic polynomial of degree d and U is a unit of $\mathcal{A}_\Gamma$. If $S = \text{supp}_t(A - P) \subseteq \Gamma_{>0}$, both corrections are supported in $S^* \setminus \{0\}$.

Proof. Let $M = S^*$, a well-ordered monoid. Write $A = P + \sum_{\eta > 0} A_\eta t^\eta$, with $A_\eta = 0$ off S . Recursively for $\eta \in M \setminus \{0\}$, define polynomials R_η, Q_η by ordinary division by P :

$$H_\eta = A_\eta - \sum_{\substack{\alpha + \beta = \eta \\ \alpha, \beta > 0}} R_\alpha Q_\beta, \tag{10}$$

$$H_\eta = PQ_\eta + R_\eta, \quad \deg R_\eta < d. \tag{11}$$

The sum is finite by lemma 2.1, and every index on its right is smaller than η . Thus this is a well-founded recursion on a set. Set $R = \sum R_\eta t^\eta$ and $Q = \sum Q_\eta t^\eta$. Because the remainders have a uniform degree bound, R is a finite polynomial over $\mathfrak{m}_\Gamma$, and $Q \in \mathcal{A}_\Gamma$. Equations (10)–(11) give $A = (P + R)(1 + Q)$ coefficient by coefficient, and the second factor is a unit by lemma 5.1.

For uniqueness, suppose two pairs differ. The union of their positive supports is well ordered, so there is a least exponent η at which either pair differs. At that exponent, equality of the products gives

$$P(Q_\eta - Q'_\eta) + (R_\eta - R'_\eta) = 0.$$

The second summand has degree less than d . Uniqueness of polynomial division forces both differences to vanish, a contradiction. For $d = 0$, the normalization gives $P = 1$, $W = 1$ and $U = A$. $\square$

This proof is deliberately not a claim that a Banach iteration converges; see [remark 4.4](#). It remains valid when no positive element of Γ has cofinal integer multiples. The uniformly bounded remainder degree is the mechanism that extracts a finite zero scheme from an infinite Hahn series.

Corollary 5.3 (The certificate survives enlargement of the value group). *Let $\Gamma \subseteq \Delta$ be an ordered-group inclusion and regard $A \in \mathcal{A}_\Gamma$ as an element of $\mathcal{A}_\Delta$ under the canonical inclusion. The factorization $A = WU$ of [theorem 5.2](#) computed over Γ is also the factorization computed over Δ : the same W and U , with the same support certificate $S^* \setminus \{0\}$.*

Proof. The recursion (10)–(11) is indexed by S^* , computed inside $\mathbb{C}[X]$, and refers to no element of the value group beyond the exponents already occurring in S . It therefore produces the same data over Δ , and uniqueness over Δ identifies the two factorizations. $\square$

[Corollary 5.3](#) is not a formality: it is exactly what lets the same prepared polynomial be used on both sides of a scalar extension in [theorem 10.7](#), and it is available only because the preparation was proved by a support recursion rather than by an iteration whose convergence would have to be re-established in the larger field.

Example 5.4 (A concrete preparation). Take $P = X^2 - X$, $d = 2$, and

$$A = X^2 - X + t(X^3 + 1) \in \mathcal{A}_\Gamma, \quad (12)$$

so that $S = \{1\}$ and $S^* = \mathbb{N}$. The recursion gives $Q_1 = X + 1$, $Q_2 = -1$, $Q_3 = 3$ and the first three remainders

$$R_1 = X + 1, \quad R_2 = -3X - 1, \quad R_3 = 8X + 2,$$

each of degree less than 2 as required. For instance $X^3 + 1 = (X^2 - X)(X + 1) + (X + 1)$ gives the first step, and $H_2 = -R_1Q_1 = -(X + 1)^2 = (X^2 - X)(-1) + (-3X - 1)$ the second. The recursion is run to perturbation order eight by the accompanying finite check suite ([appendix C](#)).

6 Exact root counting at every scale

6.1 Initial polynomials on a ball

For $\gamma \in \Gamma$, define the closed valuation ball

$$B_\gamma = \{z \in K_\Gamma : v(z) \geq \gamma\}.$$

A larger ball corresponds to a smaller, typically negative, γ . For $F \in \mathcal{E}_\Gamma \setminus \{0\}$, put

$$m_\gamma(F) = \min_{n: a_n \neq 0} \{v(a_n) + n\gamma\}, \quad (13)$$

$$I_\gamma(F)(X) = \sum_{v(a_n) + n\gamma = m_\gamma(F)} \text{lc}(a_n) X^n. \quad (14)$$

The minimum is attained, and attained at only finitely many indices, and the sum is a finite nonzero polynomial, precisely because the weighted coefficient valuations diverge ([theorem 3.1](#)). Equivalently $I_\gamma(F)$ is the reduction of $t^{-m_\gamma(F)} F(t^\gamma X)$. Write $d_\gamma(F) = \deg I_\gamma(F)$ and $e_\gamma(F) = \text{ord}_{X=0} I_\gamma(F)$.

A zero's multiplicity is the order of the translated formal power series $F(z + Z) \in K_\Gamma[[Z]]$; the translation identity (6) makes this intrinsic.

Corollary 6.1 (Finite zero geometry on the integral disk). *Let $A \in \mathcal{A}_\Gamma$ be nonzero with $w(A) = 0$ and initial polynomial monic of degree d after normalization, and let W be its prepared polynomial. The zeros of A in $\mathcal{O}_\Gamma$ are exactly the d roots of W with multiplicity, all of them integral, and for each $c \in \mathbb{C}$ the number of them with residue c is the multiplicity of c as a root of the initial polynomial. In particular a nonzero element of $\mathcal{A}_\Gamma$ has only finitely many zeros on the integral disk. No complex compactness is used.*

Proof. This is the content of the first two paragraphs of the proof of [theorem 6.2](#) below, applied at $\gamma = 0$. $\square$

Theorem 6.2 (Ball, shell, and residue-direction counts). *For $F \in \mathcal{E}_\Gamma \setminus \{0\}$ and $\gamma \in \Gamma$:*

- (a) B_γ contains exactly $d_\gamma(F)$ zeros of F , counted with multiplicity.
- (b) The open ball $\{z : v(z) > \gamma\}$ contains exactly $e_\gamma(F)$ zeros, and the shell $\{z : v(z) = \gamma\}$ contains $d_\gamma(F) - e_\gamma(F)$ zeros.
- (c) For each $c \in \mathbb{C}$, the zeros for which $v(z) \geq \gamma$ and $\text{res}(t^{-\gamma}z) = c$ have total multiplicity $\text{ord}_{X=c} I_\gamma(F)$.

Proof. Let c_0 be the leading coefficient of $I_\gamma(F)$ and apply [theorem 5.2](#) to

$$A = \frac{F(t^\gamma X)}{c_0 t^{m_\gamma(F)}}.$$

It gives $A = WU$ with $W \in \mathcal{O}_\Gamma[X]$ monic of degree $d_\gamma(F)$, $\text{res}(W) = I_\gamma(F)/c_0$, and U of initial polynomial 1. The field K_Γ is algebraically closed by the standard Hahn-field theorem [4, Corollary 4], so W has exactly its degree many roots in K_Γ , counted with multiplicity.

Every root x of W is integral: if $v(x) < 0$, the leading term x^d has valuation $dv(x)$, strictly smaller than that of every lower-degree term whose coefficient is integral, and such terms cannot sum to zero. On $\mathcal{O}_\Gamma$, evaluation of U has residue 1 and is nonzero. After translation by an integral root, U has nonzero constant term, so its formal local power series is a unit. Hence zeros and multiplicities of A in $\mathcal{O}_\Gamma$ coincide exactly with those of W .

Factor $W = \prod_j (X - x_j)$ and reduce modulo $\mathfrak{m}_\Gamma$. Then $\text{res}(W) = \prod_j (X - \text{res}(x_j))$, which proves the residue-direction count. The direction $c = 0$ is exactly the open ball, and the nonzero directions make up the shell. Rescale back by $Z = t^\gamma X$. $\square$

Corollary 6.3 (Units, faithfulness, and values). *A nonzero entire function without zeros is a nonzero constant. Distinct elements of $\mathcal{E}_\Gamma$ define distinct functions on K_Γ . Every nonconstant element of $\mathcal{E}_\Gamma$ takes every value in K_Γ .*

Proof. If a zero-free F has $a_0 = 0$ it vanishes at zero, a contradiction. If $a_n \neq 0$ for some $n > 0$, choose $\delta > 0$ and

$$\gamma = \frac{v(a_0) - v(a_n)}{n} - \delta.$$

Then $v(a_n) + n\gamma < v(a_0)$, so the constant coefficient is not an active term in (14) and the initial polynomial has positive degree; [theorem 6.2](#) supplies a zero. Therefore F is constant.

For faithfulness, a nonzero F has only finitely many zeros in any B_γ , whereas that ball contains infinitely many elements $t^\gamma c$ with $c \in \mathbb{C}$; apply this to the difference of two series. Finally, for any $b \in K_\Gamma$ a nonconstant $F - b$ cannot be zero-free, so F takes the value b . $\square$

Lemma 6.4 (Division by a linear zero factor). *If $F \in \mathcal{E}_\Gamma$ and $F(r) = 0$, then $F = (Z - r)G$ with $G \in \mathcal{E}_\Gamma$.*

Proof. Set $H(Z) = F(r + Z)$, which is entire by [proposition 3.5](#), and note that $H(0) = 0$. For each γ , write $H(t^\gamma X) = \sum_\eta P_\eta(X)t^\eta \in \mathcal{A}_\Gamma$. Its constant coefficient is zero, so every P_η is divisible by X . Dividing these finite polynomials by X preserves the well-ordered Hahn support and gives an element of $\mathcal{A}_\Gamma$. Consequently

$$(H/Z)(t^\gamma X) = t^{-\gamma} H(t^\gamma X)/X \in \mathcal{A}_\Gamma.$$

By [theorem 3.1\(2\)](#), H/Z is entire. Translating back gives $G(Z) = (H/Z)(Z - r) \in \mathcal{E}_\Gamma$ and $F = (Z - r)G$. $\square$

6.2 Stability of the counts

Corollary 6.5 (Initial-form stability). *If $F, G \in \mathcal{E}_\Gamma$ satisfy $w((F - G)(t^\gamma X)) > m_\gamma(F)$, then $m_\gamma(G) = m_\gamma(F)$ and $I_\gamma(G) = I_\gamma(F)$. All three counts in [theorem 6.2](#) are therefore unchanged.*

Proof. A perturbation supported strictly above the leading exponent changes no coefficient at that exponent; apply [theorem 6.2](#). $\square$

Proposition 6.6 (Locally finite Newton profile). *Let $F \in \mathcal{E}_\Gamma \setminus \{0\}$ and let $[\gamma_0, \gamma_1]$ be an interval of Γ . On that interval, $\gamma \mapsto m_\gamma(F)$ is the minimum of finitely many affine functions $\gamma \mapsto v(a_n) + n\gamma$. The set of active indices changes only at the slopes*

$$\frac{v(a_i) - v(a_j)}{j - i} \in \Gamma \quad (i < j),$$

and between two consecutive such values the three counts of [theorem 6.2](#) are constant.

Proof. By [theorem 3.1\(4\)](#) applied at γ_0 and at γ_1 , only finitely many indices n can attain the minimum (13) for some γ in the interval: all but finitely many satisfy $v(a_n) + n\gamma > m_{\gamma_0}(F) + |m_{\gamma_1}(F) - m_{\gamma_0}(F)|$ throughout it. A minimum of finitely many affine functions is piecewise affine, and two of them exchange dominance exactly at the displayed slope, which lies in Γ by divisibility. [Corollary 6.5](#) gives the constancy. $\square$

This is a valuation statement, a non-Archimedean analogue of a Rouché stability argument; it is not a theorem about Euclidean circles in the fine surcomplex topology. Its purpose here is to retain exact multiplicities in the factorization argument of [section 7](#).

7 Radial divisors, canonical products and factorization

7.1 Which zero sets are admissible?

Definition 7.1. An effective divisor on K_Γ is a function $D : K_\Gamma \rightarrow \mathbb{N}$ with set-sized support. It is *radially finite*, equivalently *ball-finite*, if

$$\sum_{z \in B_\gamma} D(z) < \infty \quad (\gamma \in \Gamma).$$

The sum means a finite total multiplicity, not a Hahn sum. Let $\text{Div}_{\text{rad}}^+(K_\Gamma)$ denote the monoid of these divisors.

The two source manuscripts used the two displayed adjectives for this one condition; we keep *radially finite* and record *ball-finite* as the synonym. This is a restriction on valuation balls, not on discrete subsets for the fine surreal topology, and the distinction is necessary: fine discreteness alone would impose no useful control on an infinite collection inside a fixed bounded valuation ball. Radial finiteness is exactly what the root count of [theorem 6.2](#) supplies.

Lemma 7.2 (Countability and escape). *An infinite radially finite divisor has countably many points counted with multiplicity. Its nonzero points can be enumerated as $(\rho_j)_{j \geq 1}$, with repetitions for multiplicities, and*

$$v(\rho_j) \longrightarrow -\infty \quad \text{coinitially in } \Gamma.$$

In particular an infinite radially finite divisor forces $\text{cf}(\Gamma) = \aleph_0$; and when $\text{cf}(\Gamma) > \aleph_0$, every radially finite divisor is finite.

Proof. Choose countably many distinct occurrences from the infinite divisor. Their valuations cannot have a common lower bound, for otherwise one ball would contain infinitely many occurrences. They form a countable coinitial subset of Γ , whose negatives are a countable cofinal subset. Balls at those scales exhaust K_Γ and each contains only finitely many divisor occurrences, so their union contains the whole divisor and is countable. In any one-to-one enumeration of its occurrences, only finitely many indices have valuation at least any fixed γ , which is the asserted escape. The occurrence at zero, if present, has finite multiplicity and is removed before enumerating the nonzero points. $\square$

7.2 Construction and exact factorization

Theorem 7.3 (Canonical product for a radial divisor). *Let $D \in \text{Div}_{\text{rad}}^+(K_\Gamma)$, let $m = D(0)$, and enumerate its nonzero occurrences as ρ_j . The product*

$$P_D(Z) = Z^m \prod_j \left(1 - \frac{Z}{\rho_j}\right) \tag{15}$$

is a well-defined element of $\mathcal{E}_\Gamma$, independent of the enumeration. Its divisor is exactly D , and the coefficient of Z^m is 1. The product is finite when the nonzero divisor is finite. Its partial products converge to it in the weighted Gauss valuation at every scale. No exponential correction factors are required: the theory is genus zero at every rank.

Proof. Only the infinite case needs a support argument. Put $b_j = \rho_j^{-1}$. Fix $\gamma \in \Gamma$ and substitute $Z = t^\gamma X$. By [lemma 7.2](#), $v(b_j) + \gamma \rightarrow +\infty$. There are only finitely many factors for which $v(b_j) + \gamma \leq 0$; set them aside. For the rest, the supports of $b_j t^\gamma$ have a well-ordered positive union and are locally finite, by [lemma 2.2](#).

Expand the tail product over finite subsets of indices. Every exponent belongs to the monoid generated by that positive union. For a fixed exponent, [lemma 2.1](#) gives finitely many words in the support, and local finiteness gives finitely many choices of factor labels carrying those exponents. Thus only finitely many product terms contribute, and the coefficient at that exponent is a polynomial in X . The tail product is an element of $\mathcal{A}_\Gamma$ with initial polynomial 1, hence a unit. Multiplying by the finitely many set-aside factors and by $t^{m\gamma} X^m$ gives an element of $\mathcal{A}_\Gamma$; the same estimate shows that the finite partial products approach it in m_γ .

For each fixed ordinary degree, this construction sums the same finite-subset coefficient family multiplied by $t^{k\gamma}$. Consequently the elements obtained for different γ are rescalings of one series in

$K_\Gamma[[Z]]$, which belongs to $\mathcal{E}_\Gamma$ by [theorem 3.1\(2\)](#). Strong regrouping also proves independence of the enumeration.

On B_γ , the set-aside factors give precisely the roots in that ball, while the remaining unit is nowhere zero on it. Local translation shows that multiplicities are the displayed factor multiplicities. Since the balls cover K_Γ , the divisor is exactly D . Every factor of the nonzero-root product has constant term 1, so the coefficient of Z^m is 1. $\square$

Theorem 7.4 (Factorization by zeros). *Let $F = \sum a_n Z^n \in \mathcal{E}_\Gamma \setminus \{0\}$ and let $m = \text{ord}_{Z=0} F$. Its zero divisor is radially finite. If ρ_j are all its nonzero zeros, repeated with their multiplicities, then*

$$F(Z) = a_m Z^m \prod_j \left(1 - \frac{Z}{\rho_j}\right). \quad (16)$$

Thus the zero divisor determines F up to a nonzero scalar.

Proof. Radial finiteness is [theorem 6.2](#). Let $G = P_{\text{Div}(F)}$ be the canonical product from [theorem 7.3](#). Since F and G have the same order m at zero, the formal quotient

$$Q = \frac{F/Z^m}{G/Z^m} \in K_\Gamma[[Z]]$$

exists, because G/Z^m has constant coefficient 1.

We claim $Q \in \mathcal{E}_\Gamma$. Fix γ . Preparation of $F(t^\gamma X)$ and $G(t^\gamma X)$ writes them as

$$F(t^\gamma X) = c_F W_F U_F, \quad G(t^\gamma X) = c_G W_G U_G,$$

with $c_F, c_G \in K_\Gamma^\times$, units U_F, U_G of $\mathcal{A}_\Gamma$, and monic W_F, W_G all of whose roots lie in $\mathcal{O}_\Gamma$. Those roots are exactly the normalized roots of F and of G in B_γ , with multiplicity, so $W_F = W_G$. Cancellation in $K_\Gamma[[X]]$, including the common factor X^m , gives

$$Q(t^\gamma X) = \frac{c_F}{c_G} \frac{U_F}{U_G} \in \mathcal{A}_\Gamma.$$

This holds at every scale, so [theorem 3.1\(2\)](#) proves the claim.

The identity $F = GQ$ is now an identity of entire functions. Orders at any translated centre add in the formal power-series ring. Since F and G have identical zero multiplicities, Q has no zeros, so Q is constant by [corollary 6.3](#). Comparing the coefficient of Z^m gives $Q = a_m$. $\square$

The quotient step is important and is the reason the preparation theorem was proved at *every* scale. Equal zero sets alone do not justify dividing two infinite series in the entire category, and no compactness or transfinite convergence assertion replaces that argument. Note also the order of the argument: the quotient is shown strongly evaluable at all scales *before* [theorem 3.1](#) upgrades it to an entire, valuation-convergent series. In particular the proof never assumes that a zero-free restricted series is restricted-invertible — by [example 4.3](#) that is exactly what can fail in higher rank.

Corollary 7.5 (Divisibility, GCDs, and least common multiples). *The map $F \mapsto \text{Div}(F)$ induces an isomorphism of commutative monoids*

$$(\mathcal{E}_\Gamma \setminus \{0\})/K_\Gamma^\times \cong \text{Div}_{\text{rad}}^+(K_\Gamma).$$

For nonzero $F, G \in \mathcal{E}_\Gamma$, G divides F in $\mathcal{E}_\Gamma$ if and only if $\text{Div}(G) \leq \text{Div}(F)$ pointwise. Greatest common divisors and least common multiples are given by the pointwise minimum and maximum of divisors. In particular $\mathcal{E}_\Gamma$ is a GCD domain.

Proof. Surjectivity is [theorem 7.3](#); injectivity modulo constants is [theorem 7.4](#). Divisors add under products by Taylor expansion. If the divisor of G is contained in that of F , their difference is radially finite, so its canonical product supplies the quotient, up to the scalar fixed by [theorem 7.4](#). Pointwise minima and maxima of two radial divisors remain radially finite; their canonical products therefore have the defining universal divisibility properties of a GCD and an LCM. $\square$

Corollary 7.6 (Finite jets and evaluation ideals). *For $x \in K_\Gamma$ and $r \geq 1$, the kernel of the Taylor jet map*

$$\mathcal{E}_\Gamma \longrightarrow K_\Gamma[U]/(U^r), \quad F \longmapsto F(x + U) \bmod U^r$$

is $(Z - x)^r \mathcal{E}_\Gamma$, and the map is surjective. In particular evaluation at x has kernel $(Z - x)\mathcal{E}_\Gamma$ and residue field K_Γ .

Proof. Vanishing of the jet is exactly a zero of order at least r ; apply [corollary 7.5](#). Every jet is realized by a finite polynomial in $Z - x$. $\square$

Corollary 7.7 (Rigidity and value distribution). *Let $F \in \mathcal{E}_\Gamma$ be nonzero.*

- (a) *F has finitely many zeros if and only if F is a polynomial; a zero-free F is a nonzero constant.*
- (b) *If F is nonconstant it is surjective onto K_Γ ; if F is nonpolynomial, every fibre $F^{-1}(b)$ is infinite.*
- (c) *If F is injective as a function on K_Γ , then F is affine of degree one.*

Proof. (a) A polynomial has finitely many zeros, and conversely (16) with a finite product is a polynomial. Zero-freeness is [corollary 6.3](#). (b) Surjectivity is [corollary 6.3](#); if F is nonpolynomial then $F - b$ is nonpolynomial, so by (a) it has infinitely many zeros. (c) An injective F is nonconstant, hence a polynomial by (b), say of degree $d \geq 1$ with leading coefficient a_d . For each b , injectivity makes the divisor of $F - b$ a single point r_b with multiplicity d , so $F(Z) - b = a_d(Z - r_b)^d$ by [theorem 7.4](#). Comparing two values $b \neq b'$ gives $a_d[(Z - r_b)^d - (Z - r_{b'})^d] = b' - b$, a nonzero constant. For $d \geq 2$ the left side has a nonzero coefficient of Z^{d-1} unless $r_b = r_{b'}$, and $r_b = r_{b'}$ makes it zero; either way we contradict $b \neq b'$. Hence $d = 1$. $\square$

Part (b) and part (c) are familiar non-Archimedean value distribution, stated here for completeness at arbitrary rank and not offered as independent priority claims.

Corollary 7.8 (Conjugation and real-coefficient functions). *A nonzero entire series has all coefficients in $\mathbb{R}((t^\Gamma))$ if and only if its zero divisor is invariant under coefficientwise conjugation and its first nonzero coefficient lies in $\mathbb{R}((t^\Gamma))$.*

Proof. Conjugating coefficients conjugates each value and each zero with its multiplicity. Conversely, under the stated conditions the conjugate series has the same divisor and the same first nonzero coefficient, so uniqueness in [theorem 7.4](#) makes the two series equal. $\square$

7.3 What “factorization” does not imply

Proposition 7.9. *If $\mathcal{E}_\Gamma$ contains a nonpolynomial function, it is not Noetherian and is not atomic. Its irreducible elements are precisely the scalar multiples of linear polynomials.*

Proof. By [theorem 7.4](#), a nonpolynomial entire function has infinitely many zeros with multiplicity. Let $\rho_1, \rho_2, \dots$ be an escaping sequence of nonzero roots and put $T_n(Z) = \prod_{j \geq n} (1 - Z/\rho_j)$. Then $T_n = (1 - Z/\rho_n)T_{n+1}$, so

$$(T_1) \subsetneq (T_2) \subsetneq (T_3) \subsetneq \dots,$$

strictness following either by evaluating multiplicities at ρ_n or from [corollary 7.5](#). Thus the ring is not Noetherian.

A nonunit has a zero. If its divisor has at least two occurrences, separating one occurrence from the remaining divisor gives a product of two nonunits. A singleton divisor gives a scalar multiple of one linear factor and is irreducible. Consequently a function with an infinite divisor cannot be a *finite* product of irreducibles, which is the failure of atomicity. $\square$

The canonical product theorem is therefore not a unique-factorization-domain theorem. Its product is infinite in a precisely controlled Hahn sense; an algebraic UFD requires finite irreducible factorizations.

8 The higher-rank obstruction to interpolation and Bézout

8.1 A sequence of genuinely new Archimedean scales

For positive $\alpha, \beta \in \Gamma$, write

$$\beta \gg_A \alpha \iff \beta > k\alpha \text{ for every integer } k \geq 1.$$

This is Archimedean domination: a genuinely new Archimedean class, not merely a large finite ratio.

Lemma 8.1. *Suppose $\text{cf}(\Gamma) = \aleph_0$ and Γ has no order unit. There is a positive cofinal sequence $(\gamma_n)_{n \geq 1}$ with $\gamma_{n+1} \gg_A \gamma_n$ for every n .*

Proof. Choose a positive cofinal sequence (q_n) . After selecting γ_n , the set $\{k\gamma_n : k \geq 1\}$ is not cofinal, since γ_n is not an order unit; hence it has a strict upper bound $b_n \in \Gamma$. Choose $\gamma_{n+1} > \max\{b_n, q_{n+1}, \gamma_n\}$ and start with $\gamma_1 \geq q_1 > 0$. The resulting sequence has the required domination and is cofinal because it dominates the q_n . $\square$

8.2 A universal bound on values at escaping monomials

The next lemma is the core obstruction. It is short, but it uses both halves of [definition 1.1](#): a common lower bound on the coefficient supports, and legitimate evaluation at the large argument.

Lemma 8.2 (Archimedean growth barrier). *Let $\gamma_n > 0$ tend cofinally to $+\infty$ (for example, let (γ_n) be increasing and cofinal), and let $\lambda_n > 0$ satisfy $\lambda_n \gg_A \gamma_n$ for every sufficiently large n . For each $H \in \mathcal{E}_\Gamma$, eventually*

$$v(H(t^{-\gamma_n})) > -\lambda_n. \tag{17}$$

The value $+\infty$ is allowed when H vanishes at the argument.

Proof. The assertion is immediate for $H = 0$. Write $H = \sum_{k \geq 0} h_k Z^k$ and let $m = \min_{h_k \neq 0} v(h_k)$, which exists by strong summability at $Z = 1$. For large n , $\gamma_n > |m|$. Then each nonzero term at $Z = t^{-\gamma_n}$ satisfies

$$v(h_k t^{-k\gamma_n}) \geq m - k\gamma_n > -(k+1)\gamma_n > -\lambda_n.$$

The terms' valuations tend cofinally to infinity as $k \rightarrow \infty$ by [theorem 3.1](#), and their nonempty set has a minimum attained at a finite k . That minimum is strictly greater than $-\lambda_n$, and the valuation of their Hahn sum is at least the minimum, giving [\(17\)](#). $\square$

The eventual condition on γ_n cannot be replaced by cofinality of its range. In the reverse-lexicographic group $\bigoplus_{j \geq 0} \mathbb{Q}e_j$, set $\gamma_{2n} = e_{n+2}$, $\gamma_{2n+1} = e_0$ and $\lambda_{2n} = e_{n+3}$, $\lambda_{2n+1} = e_1$ for $n \geq 0$. The range of (γ_n) is cofinal and $\lambda_n \gg_A \gamma_n$ always, but the constant entire function $H = t^{-e_2}$ violates the conclusion at every odd index. All applications below use the increasing sequence of [lemma 8.1](#).

Corollary 8.3 (Explicit failure of value interpolation). *For the sequence of [lemma 8.1](#), the prescription*

$$H(t^{-\gamma_n}) = t^{-\gamma_{n+1}} \quad (n \geq 1)$$

has no solution $H \in \mathcal{E}_\Gamma$. Each finite subcollection of these prescriptions has a polynomial solution.

Proof. The valuations of the prescribed values are $-\gamma_{n+1}$, in conflict with [lemma 8.2](#). The nodes are distinct, so ordinary finite Lagrange interpolation over the field K_Γ realizes any finite subcollection. $\square$

This failure is not caused by nodes accumulating in a bounded ball: their divisor is radially finite and is therefore the divisor of an entire function by [theorem 7.3](#). Realizability of zeros is strictly weaker than realizability of values.

8.3 Two functions with no common zero and no Bézout identity

Theorem 8.4 (An explicit pointwise Bézout failure). *Assume the hypotheses of [lemma 8.1](#) and choose its sequence (γ_n) . Define*

$$\rho_n = t^{-\gamma_n}, \quad \sigma_n = t^{-\gamma_n} + t^{\gamma_{n+1}}, \quad (18)$$

$$f(Z) = \prod_{n \geq 1} (1 - Z/\rho_n), \quad g(Z) = \prod_{n \geq 1} (1 - Z/\sigma_n). \quad (19)$$

Then $f, g \in \mathcal{E}_\Gamma$, both have constant coefficient 1, their roots are simple and disjoint, and

$$\gcd(f, g) = 1, \quad (f, g) \neq \mathcal{E}_\Gamma.$$

In particular there are no $A, B \in \mathcal{E}_\Gamma$ with $Af + Bg = 1$.

Proof. We have $v(\rho_n) = v(\sigma_n) = -\gamma_n$, so both root sequences escape every valuation ball and [theorem 7.3](#) defines the two entire functions. Different indices have different valuations; at the same index $\sigma_n - \rho_n = t^{\gamma_{n+1}} \neq 0$. Their zero sets are consequently disjoint and simple, and [corollary 7.5](#) gives their GCD.

We compute $v(g(\rho_n))$ exactly. For $j < n$, ρ_n/σ_j has negative valuation, so

$$v(1 - \rho_n/\sigma_j) = \gamma_j - \gamma_n.$$

For $j = n$,

$$1 - \rho_n/\sigma_n = \frac{\sigma_n - \rho_n}{\sigma_n}, \quad v(1 - \rho_n/\sigma_n) = \gamma_{n+1} + \gamma_n.$$

For $j > n$ the ratio has positive valuation, and the infinite tail product has residue 1 and valuation zero by the positive-support product argument used in [theorem 7.3](#). Therefore

$$\lambda_n := v(g(\rho_n)) = \gamma_{n+1} + (2 - n)\gamma_n + \sum_{j < n} \gamma_j. \quad (20)$$

The sum here is finite. Since $0 < \gamma_j < \gamma_n$ for $j < n$, the part of (20) other than γ_{n+1} is bounded in absolute value by a finite multiple of γ_n . Hence $\lambda_n \gg_A \gamma_n$.

Suppose $Af + Bg = 1$. Evaluate at ρ_n . The values are well defined and evaluation is multiplicative, so

$$B(\rho_n) = \frac{1}{g(\rho_n)}, \quad v(B(\rho_n)) = -\lambda_n.$$

For all sufficiently large n this contradicts [lemma 8.2](#). Thus the ideal is proper. $\square$

The cancellation-sensitive step is (20). No unproved assertion that nearby zeros force a small value is used: the contribution of each earlier root is retained, and the infinite tail is proved to be a unit. The earlier-root terms can lower the valuation by finite multiples of γ_n , but cannot offset the new Archimedean scale γ_{n+1} .

Corollary 8.5 (A finitely generated ideal invisible to point tests). *In the situation of [theorem 8.4](#) there is a proper two-generated ideal contained in no evaluation maximal ideal $(Z - x)\mathcal{E}_\Gamma$. Every maximal ideal containing it is therefore not the kernel of evaluation at any $x \in K_\Gamma$.*

Proof. The ideal (f, g) is proper but f and g do not vanish simultaneously. By [corollary 7.6](#), containment in $(Z - x)\mathcal{E}_\Gamma$ would mean that both vanish at x . A maximal ideal containing a proper ideal exists by Zorn's lemma; none of these maximal ideals is an evaluation ideal. $\square$

This is a failure of the assertion that *every finitely generated proper ideal has a common zero*. We do not call it a metric corona counterexample, because no norm lower bound and no bounded-function algebra has been specified. Nor do we claim that non-evaluation maximal ideals are absent in the rank-one entire ring; the statement concerns a *two-generated* ideal already lacking a common zero.

8.4 The missing image of the infinite jet map

Let D be the simple divisor with nodes ρ_n . The evaluation map

$$\text{ev}_D : \mathcal{E}_\Gamma \longrightarrow \prod_{n \geq 1} K_\Gamma, \quad H \longmapsto (H(\rho_n))_n$$

has kernel $f\mathcal{E}_\Gamma$ by [corollary 7.5](#), and it is not surjective by [corollary 8.3](#). In particular

$$\mathcal{E}_\Gamma / (f) \hookrightarrow \prod_{n \geq 1} K_\Gamma$$

is a proper subring. The element $g \bmod f$ maps to an everywhere nonzero sequence, hence to a unit of the product ring, but it is not a unit in $\mathcal{E}_\Gamma / (f)$: an inverse there would give representatives satisfying a Bézout identity.

Thus the elementary implication “invertible at each node implies invertible in the global quotient” is exactly where unrestricted infinite Chinese remainder reasoning fails. Every finite projection of this map is surjective. Neither a finite truncation nor a finite root calculation can detect the global failure; see [appendix C](#).

9 The positive case: an order unit restores interpolation

9.1 A rank-one coarsening with the same entire functions

It would be incorrect to infer that the preceding obstruction occurs for every higher-rank group. A group may have several convex levels and still have a greatest Archimedean class. In that case a single cofinal scale permits ordinary non-Archimedean analytic methods.

Lemma 9.1 (Real-valued coarsening). *Suppose $h > 0$ is an order unit of Γ . There is an order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$, whose kernel is the convex subgroup*

$$H = \{\gamma \in \Gamma : n|\gamma| < h \text{ for all } n \geq 1\}.$$

The formula

$$|a|_\pi = \exp(-\pi(v(a))) \quad (a \neq 0), \quad |0|_\pi = 0$$

defines a nontrivial real-valued non-Archimedean absolute value on K_Γ . Its topology agrees with the intrinsic valuation topology, and K_Γ is complete for it. Furthermore

$$\mathcal{E}_\Gamma = \left\{ \sum_{n \geq 0} a_n Z^n : |a_n|_\pi R^n \longrightarrow 0 \text{ for every real } R > 0 \right\}. \quad (21)$$

Proof. Being an order unit bounds every γ above and below by integer multiples of h . Define

$$\pi(\gamma) = \sup\{q \in \mathbb{Q} : qh \leq \gamma\} \in \mathbb{R}.$$

Rational upper and lower approximations show $\pi(\alpha + \beta) = \pi(\alpha) + \pi(\beta)$ and $\pi(h) = 1$; monotonicity is immediate. The kernel consists exactly of the elements infinitesimal relative to h , giving the displayed H . In particular $\pi(\gamma) > n$ implies $\gamma > nh$, and $\gamma > (n+1)h$ implies $\pi(\gamma) > n$. The balls specified by integer multiples of h are cofinal among all valuation neighbourhoods of zero, and these inequalities show that the real-valued and ordered-group topologies agree.

For completeness, take a Cauchy sequence for $|\cdot|_\pi$ and choose a subsequence (x_{j_n}) with $v(x_{j_{n+1}} - x_{j_n}) > nh$. The differences have valuations tending cofinally to infinity, so their Hahn sum exists by lemma 2.2; adding it to x_{j_1} gives a limit of the subsequence, and the Cauchy property gives the same limit for the whole sequence. Algebraic closedness of K_Γ was already supplied by the Hahn-field theorem.

By theorem 3.1, the left side of (21) is characterized by $v(a_n)/n \rightarrow +\infty$ in Γ . This is equivalent to $\pi(v(a_n))/n \rightarrow +\infty$ in $\mathbb{R}$: integer multiples of h test the forward implication, and for the reverse one chooses a real bound strictly exceeding $\pi(\gamma)$ for a prescribed $\gamma \in \Gamma$. The usual elementary coefficient test identifies this last condition with the right side of (21): explicitly, it implies $\pi(v(a_n)) - n \log R \rightarrow +\infty$ for every R ; conversely, failure of the ratio condition gives infinitely many ratios at most some C , and then $R = e^{C+1}$ violates the coefficient test. $\square$

This is a *coarsening*, not a valuation-preserving embedding of Γ in $\mathbb{R}$. The kernel H may be nonzero, and residue-direction data is not identified across the two valuations: the coarsening changes residue information even while preserving the topology and the entire ring. It is the cofinality of the topology and the all-radius coefficient test that permit its use here.

9.2 A complete Hermite interpolation argument

The next proof records the damping mechanism that fails in [section 8](#). It is included instead of importing an unqualified analytic interpolation theorem.

Theorem 9.2 (Arbitrary radial Hermite interpolation). *Assume Γ has an order unit. Let (x_n) be distinct points of a radially finite set, and assign a positive integer m_n to each point so that the divisor $\sum_n m_n[x_n]$ is radially finite. For arbitrary jets $J_n \in K_\Gamma[U]/(U^{m_n})$ there exists $F \in \mathcal{E}_\Gamma$ with*

$$F(x_n + U) \equiv J_n(U) \pmod{U^{m_n}} \quad \text{for every } n.$$

Proof. Use $|\cdot| = |\cdot|_\pi$ from [lemma 9.1](#). A radially finite infinite set can be enumerated with $|x_n| \rightarrow \infty$: a ball of any fixed real absolute-value radius is contained in some original valuation ball, and conversely an original ball has bounded $|\cdot|_\pi$ radius. Handle the finitely many nodes with $|x_n| \leq 1$, including zero, first by a finite Hermite polynomial, and include their factors in every later vanishing product. Reindex the remaining nodes by $n \geq 1$, so that $|x_n| > 1$ and $|x_n| \rightarrow \infty$.

Suppose a polynomial S_{n-1} has the required jets at earlier nodes and put

$$Q_{n-1}(Z) = \prod_{j < n} (Z - x_j)^{m_j},$$

including the finitely many initially handled nodes. It is a unit in the local jet ring at x_n , so there is a unique jet

$$B_n(U) = \frac{J_n(U) - S_{n-1}(x_n + U)}{Q_{n-1}(x_n + U)} \pmod{U^{m_n}}.$$

For a positive integer N define the polynomial of degree less than m_n

$$r_{n,N}(U) = B_n(U)(1 + U/x_n)^{-N} \pmod{U^{m_n}} \quad (22)$$

and the polynomial correction

$$p_{n,N}(Z) = Q_{n-1}(Z)(Z/x_n)^N r_{n,N}(Z - x_n). \quad (23)$$

It vanishes to the required orders at all earlier nodes, and by (22) its jet at x_n is the required residual. The parameter N can therefore be used purely to control its size.

Set $R_n = |x_n|^{1/2}$, so $1 < R_n < |x_n|$ and $R_n \rightarrow \infty$. For a polynomial $P = \sum p_k Z^k$ write $\|P\|_R = \max_k |p_k| R^k$. If $B_n(U) = \sum_{j < m_n} b_j U^j$, the coefficient of U^ℓ in (22) has absolute value at most $\max_{j \leq \ell} |b_j| |x_n|^{j-\ell}$. This bound is independent of N , because the binomial coefficients $\binom{-N}{k}$ are integers and so have absolute value at most 1. Since $R_n < |x_n|$, substitution $U = Z - x_n$ gives

$$\|r_{n,N}(Z - x_n)\|_{R_n} \leq \max_{j < m_n} |b_j| |x_n|^j =: C'_n,$$

and consequently

$$\|p_{n,N}\|_{R_n} \leq \|Q_{n-1}\|_{R_n} C'_n \left(\frac{R_n}{|x_n|} \right)^N. \quad (24)$$

The right side tends to zero as the ordinary integer N tends to infinity. Choose N_n so that this norm is at most 2^{-n} and put $S_n = S_{n-1} + p_{n,N_n}$.

For every fixed real $R > 0$, eventually $R \leq R_n$, so $\|p_{n,N_n}\|_R \leq 2^{-n}$. Hence (S_n) is Cauchy in the Gauss norm at every radius. The space of series satisfying $|a_k| R^k \rightarrow 0$ is complete for that

norm: coefficientwise limits exist by completeness of K_Γ , and uniform approximation by a finite polynomial bounds the tail of the limit. Thus the limits for all radii are one common coefficient series F satisfying (21), so $F \in \mathcal{E}_\Gamma$.

Finally, every specified finite jet is preserved. For a fixed node x and order m , choose $R > |x|$. Coefficients of the translated polynomial up to degree $m - 1$ are continuous for $\|\cdot\|_R$, by the finite binomial formula and its convergent tail. All sufficiently late S_n have the prescribed jet, so their limit does too. $\square$

Theorem 9.3 (Bézout in the order-unit case). *If Γ has an order unit then $\mathcal{E}_\Gamma$ is a Bézout domain. In particular, if $f, g \in \mathcal{E}_\Gamma$ have no common zero there are $A, B \in \mathcal{E}_\Gamma$ with $Af + Bg = 1$.*

Proof. First suppose f, g are nonzero with no common zero. At each zero x of f let m_x be its multiplicity. Since $g(x) \neq 0$, the formal inverse of $g(x + U)$ exists modulo U^{m_x} . By [theorem 9.2](#) choose $B \in \mathcal{E}_\Gamma$ whose jet is that inverse at every such x . Then $1 - Bg$ has a zero of order at least m_x at each zero of f , so it is divisible by f in $\mathcal{E}_\Gamma$ by [corollary 7.5](#); write $1 - Bg = Af$.

For arbitrary nonzero f, g , divide both by their GCD d , which exists by [corollary 7.5](#). The quotients have no common zero, so their ideal is the unit ideal; multiplying the resulting identity by d shows $(f, g) = (d)$. Cases involving zero are immediate. Induct on the finite number of generators to obtain the Bézout property. $\square$

For a radially finite divisor D the full jet map is therefore surjective:

$$\mathcal{E}_\Gamma / (P_D) \cong \prod_{x \in \text{supp } D} K_\Gamma[U] / (U^{D(x)}).$$

The kernel is again given by divisibility, and surjectivity is [theorem 9.2](#). This exact unrestricted Chinese remainder statement contrasts sharply with the proper image of [section 8.4](#).

10 Change of workspace: the universal scalar-extension domain

The adjective “entire” must name its field of arguments. A formal series with coefficients in one Hahn field may be entire there and fail even strong summability at a single point of a larger field. This section gives the exact boundary, and then the exact description of everything inside it.

Throughout, $\Gamma \subseteq \Delta$ are nonzero divisible ordered abelian groups with the canonical inclusion $K_\Gamma \subseteq K_\Delta$, both full Hahn fields over the *same* coefficient field $\mathbb{C}$.

10.1 The sharp criterion for remaining entire

Theorem 10.1 (Sharp change-of-workspace criterion). *With the above notation,*

$$\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = \begin{cases} \mathcal{E}_\Gamma, & \Gamma \text{ is cofinal in } \Delta, \\ K_\Gamma[Z], & \Gamma \text{ is not cofinal in } \Delta. \end{cases} \quad (25)$$

In the second case a single monomial argument of K_Δ makes every nonpolynomial series with coefficients in K_Γ fail strong summability.

Proof. If Γ is cofinal in Δ , a sequence of elements of $\Gamma \cup \{+\infty\}$ tends cofinally to infinity in Γ if and only if it does so in Δ . Apply [theorem 3.1](#) to the coefficient slopes.

If Γ is not cofinal, choose $\delta \in \Delta$ strictly larger than every element of Γ . For a nonpolynomial $F = \sum a_n Z^n \in K_\Gamma[[Z]]$ consider evaluation at $Z = t^{-\delta}$. For any two nonzero coefficients with $m > n$,

$$(v(a_m) - m\delta) - (v(a_n) - n\delta) = v(a_m) - v(a_n) - (m - n)\delta < 0,$$

since $v(a_m) - v(a_n) \in \Gamma$ and $(m - n)\delta \geq \delta$. The leading exponents of the nonzero terms therefore form an infinite strictly descending sequence, so their support union is not well ordered. No nonpolynomial series over K_Γ can be strongly entire over K_Δ . Polynomials are entire over either field, proving (25). $\square$

10.2 The convex-hull valuation ring

Definition 10.2. The convex hull of Γ in Δ is

$$H = \text{conv}_\Delta(\Gamma) = \{s \in \Delta : |s| \leq \gamma \text{ for some } \gamma \in \Gamma_{>0}\}, \quad (26)$$

a divisible convex subgroup of Δ . Coarsening v_Δ by H gives a valuation with value group Δ/H .

Lemma 10.3. *The valuation ring of that coarsening is*

$$\mathcal{D}_{\Gamma,\Delta} = \{0\} \cup \{z \neq 0 : v_\Delta(z) \geq \gamma \text{ for some } \gamma \in \Gamma\}. \quad (27)$$

Its maximal ideal consists of zero together with the elements satisfying $v_\Delta(z) > \gamma$ for every $\gamma \in \Gamma$. Its residue field is canonically $\mathbb{C}((t^H))$. Moreover $\mathcal{D}_{\Gamma,\Delta} = K_\Delta$ exactly when Γ is cofinal in Δ .

Proof. A value s has nonnegative class modulo H exactly when it is bounded below by some member of H , and every member of H is itself bounded below by a member of Γ ; this proves (27). The class is strictly positive exactly when $s > H$, equivalently $s > \Gamma$.

For an element of this valuation ring, discard all support exponents outside H . No negative coset can occur, and the discarded positive cosets form the coarsened maximal ideal. The remaining Hahn series has support in H . This is a surjective residue homomorphism onto $\mathbb{C}((t^H))$ with the stated kernel. Finally the ring is the whole field exactly when $H = \Delta$, which by (26) is equivalent to cofinality of Γ in Δ . $\square$

The residue field here is generally *not* $\mathbb{C}$. It records all scales in the convex hull H , whereas the original fine valuation's residue field records only the exponent zero.

10.3 Exact strong evaluation, not merely a sufficient domain

Theorem 10.4 (Universal scalar-extension domain). *Let $f = \sum a_n Z^n \in \mathcal{E}_\Gamma$ be nonpolynomial. In K_Δ the family $(a_n z^n)_n$ is strongly summable if and only if $z \in \mathcal{D}_{\Gamma,\Delta}$. The domain is therefore independent of the choice of nonpolynomial f , and the values on it again lie in $\mathcal{D}_{\Gamma,\Delta}$.*

Proof. Suppose $z \neq 0$ lies in the stated ring. Choose $\gamma \in \Gamma$ with $v_\Delta(z) \geq \gamma$ and put $u = t^{-\gamma} z \in \mathcal{O}_\Delta$. Since f is entire over K_Γ , theorem 3.1(2) gives $f(t^\gamma X) \in \mathcal{A}_\Gamma \subseteq \mathcal{A}_\Delta$, and lemma 5.1 gives strong summability of its evaluation family at the integral point u . That family is the original family $(a_n z^n)_n$. With $\mu = m_\gamma(f) \in \Gamma$, the same lemma gives $v_\Delta(f(z)) \geq \mu$ unless the value is zero, so the value lies in $\mathcal{D}_{\Gamma,\Delta}$. At $z = 0$ the assertion is immediate.

Conversely suppose $s = v_\Delta(z) < \gamma$ for every $\gamma \in \Gamma$. List the infinitely many nonzero coefficients at increasing indices $n_1 < n_2 < \dots$ and write $\alpha_j = v(a_{n_j}) \in \Gamma$. Since $s < 0$,

$$\begin{aligned} (\alpha_{j+1} + n_{j+1}s) - (\alpha_j + n_js) &= \alpha_{j+1} - \alpha_j + (n_{j+1} - n_j)s \\ &\leq \alpha_{j+1} - \alpha_j + s < 0, \end{aligned}$$

the final inequality because s is below the element $\alpha_j - \alpha_{j+1}$ of Γ . These are actual leading exponents of the nonzero summands $a_{n_j}z^{n_j}$, regardless of the higher terms of z . Their strict descent prevents the support union from being well ordered, so the family is not strongly summable. $\square$

Corollary 10.5 (Cofinal extensions are exactly the entire extensions). *For nonpolynomial $f \in \mathcal{E}_\Gamma$, the same coefficient series belongs to $\mathcal{E}_\Delta$ if and only if Γ is cofinal in Δ . Polynomials always survive.*

Proof. Combine [theorem 10.4](#) with [lemma 10.3](#); polynomial evaluations are finite sums. $\square$

Remark 10.6 (The negative and the positive statement are compatible). [Theorem 10.1](#) and [theorem 10.4](#) must be read together, and [remark 1.7](#) already stated how. The first says that in the noncofinal case a nonpolynomial f is *not entire over K_Δ* , so that $\mathcal{E}_\Delta \cap K_\Gamma[[Z]]$ collapses to $K_\Gamma[Z]$; the second says that the very same series remains strongly summable on the proper subring $\mathcal{D}_{\Gamma,\Delta}$, with values in $\mathcal{D}_{\Gamma,\Delta}$, and [theorem 10.7](#) below says it keeps every one of its zeros there. There is no disagreement about whether the function survives: it survives on $\mathcal{D}_{\Gamma,\Delta}$ and nowhere else, and “not entire over K_Δ ” is precisely the statement $\mathcal{D}_{\Gamma,\Delta} \neq K_\Delta$. Indeed the noncofinal half of [theorem 10.1](#) is the special case $z = t^{-\delta}$ of the necessity half of [theorem 10.4](#), and its independent descending-exponent proof is retained because it exhibits the single witnessing point explicitly.

A cofinal extension need not preserve rank, and a noncofinal one need not introduce any algebraic element: the obstruction lies entirely in the ordered hierarchy of scales. Nor does a strongly defined value in the larger field have to be a limit there. If Δ contains a positive value above all of Γ , the original cofinal coefficient valuations in Γ no longer tend to infinity in Δ .

10.4 No new zeros, even for an infinite divisor

Theorem 10.7 (Zero conservation under arbitrary enlargement). *Let $0 \neq f \in \mathcal{E}_\Gamma$. Every zero of its strongly defined extension on $\mathcal{D}_{\Gamma,\Delta}$ belongs to K_Γ , with unchanged multiplicity. This remains true when f has infinitely many zeros.*

Proof. Suppose $z \in \mathcal{D}_{\Gamma,\Delta}$ and $f(z) = 0$. Choose $\gamma \in \Gamma$ with $v_\Delta(z) \geq \gamma$. Preparation of $f(t^\gamma X)$ over K_Γ ([theorem 5.2](#)) gives

$$f(t^\gamma X) = c t^\mu W_\gamma(X) U_\gamma(X),$$

with W_γ monic over $\mathcal{O}_\Gamma$ and U_γ a unit of $\mathcal{A}_\Gamma$ with initial polynomial 1. By [corollary 5.3](#) this is also the preparation over Δ , and by [lemma 5.1](#) the unit factor is nonzero at every point of $\mathcal{O}_\Delta$. Therefore $W_\gamma(t^{-\gamma}z) = 0$. The polynomial W_γ already splits into linear factors over the algebraically closed field K_Γ , so $t^{-\gamma}z$, and hence z , lies in K_Γ .

The same prepared polynomial has the same linear-factor multiplicities in both fields, and the evaluated unit is nonzero in both, so local orders of vanishing are unchanged. The argument uses only the finitely many zeros in one containing ball and applies separately to each zero; it does not require the global divisor to be finite. $\square$

This is the one statement of the section that imports algebraic closedness. [Appendix A](#) records that dependency.

10.5 No alternative entire continuation

Proposition 10.8 (Identity on constants). *Two formal series over K_Δ lying in $\mathcal{A}_\Delta$ and agreeing at infinitely many distinct points of $\mathcal{O}_\Delta$ are equal as formal series. In particular agreement at all ordinary complex constants suffices.*

Proof. Their difference A lies in $\mathcal{A}_\Delta$. If it were nonzero, multiplication by a nonzero scalar would normalize its Gauss value to zero and its initial polynomial to monic. By [theorem 5.2](#), the normalized series is WU with W a nonzero finite-degree polynomial and U a unit. Evaluation on $\mathcal{O}_\Delta$ is a ring homomorphism by [lemma 5.1](#), so U never vanishes there. Thus A has at most $\deg W$ distinct integral zeros, a contradiction. This finite upper bound uses no splitting of W and hence no algebraic-closedness hypothesis. $\square$

Theorem 10.9 (Obstruction to any entire-series continuation). *Suppose Γ is not cofinal in Δ and $f \in \mathcal{E}_\Gamma$ is nonpolynomial. There is no $g \in \mathcal{E}_\Delta$ agreeing with f at all points of K_Γ . In fact there is no such g agreeing with f merely at all points of $\mathbb{C}$.*

Proof. The coefficient series of f belongs to $\mathcal{A}_\Delta$, since its strong support certificate at scale zero over Γ remains valid under an ordered embedding; the same is true of g . Agreement on $\mathbb{C}$ would force the two coefficient series to be identical by [proposition 10.8](#). But [corollary 10.5](#) says that this coefficient series is not entire over K_Δ , a contradiction. $\square$

This rules out an alternative entire *power-series* continuation. It does not rule out arbitrary functions extending the pointwise values, or continuations in a different analytic category, and it asserts no density of $\mathbb{C}$ in any fine surreal topology.

10.6 The boundary is intrinsic to the function

Corollary 10.10 (Intrinsic description of the extension boundary). *Let $f \in \mathcal{E}_\Gamma$ be nonpolynomial. In any larger ordered group Δ the following three subsets generate the same convex subgroup:*

$$\Gamma, \quad \{v(a_n)/n : n \geq 1, a_n \neq 0\}, \quad \{-v(r) : r \neq 0, f(r) = 0\}.$$

Consequently the universal domain of [theorem 10.4](#) can be recovered either from the coefficient slopes alone or from the zero divisor alone.

Proof. All the displayed values lie in Γ . The slopes are cofinal in Γ by [theorem 3.1](#). There are infinitely many zeros by [corollary 7.7\(a\)](#), and their negated valuations are cofinal by [lemma 7.2](#). A cofinal subset of an ordered group generates its full convex hull in any containing ordered group. $\square$

This makes the extension obstruction intrinsic. It is not an artefact of having embedded the coefficients in an unnecessarily large workspace whose extra scales the function never detects.

10.7 Consequences for the Bézout counterexample and for $\text{No}[i]$

Corollary 10.11 (No repair of the counterexample by a Hahn extension). *Let f, g be the pair of [theorem 8.4](#). If Γ is cofinal in Δ then $f, g \in \mathcal{E}_\Delta$ and still no $A, B \in \mathcal{E}_\Delta$ satisfy $Af + Bg = 1$. If Γ is not cofinal in Δ then neither f nor g is entire on K_Δ , although both remain strongly defined on $\mathcal{D}_{\Gamma, \Delta}$ with all their zeros ([theorems 10.4](#) and [10.7](#)).*

Proof. In the cofinal case (γ_n) remains cofinal in Δ and $\gamma_{n+1} \gg_A \gamma_n$ is unchanged, so the evaluation formula (20) is unchanged. The proof of lemma 8.2 now applies to any candidate $B \in \mathcal{E}_\Delta$, even with coefficients outside K_Γ , giving the same contradiction. In the noncofinal case apply theorem 10.1 to the two nonpolynomial series, and theorems 10.4 and 10.7 for the positive half. $\square$

Corollary 10.12 (Whole-class rigidity on the surcomplex numbers). *Let $F(Z) = \sum_{n \geq 0} a_n Z^n$ be an ordinary power series with coefficients in $\mathbf{No}[i]$. If the family $(a_n z^n)_n$ is strongly normal-form-summable for every $z \in \mathbf{No}[i]$, then only finitely many a_n are nonzero.*

Proof. The union of the normal-form exponent supports of the real and imaginary parts of the countably many coefficients is a *set*. Its divisible additive span together with 1, with signs chosen to match $t^\gamma = \omega^{-\gamma}$, is a nonzero set-sized ordered subgroup $\Gamma \subseteq \mathbf{No}$, and all a_n lie in K_Γ under (1); the hypothesis makes F entire over K_Γ . Every set of surreal numbers has a surreal strict upper bound: the set-sized cut $\delta = \{ \Gamma \cup \{0\} \mid \emptyset \}$ supplies a positive $\delta \in \mathbf{No}$ larger than every member of Γ . Enlarge Γ to its divisible span with δ . If F were nonpolynomial, the descending-leading-exponent argument of theorem 10.1 — equivalently theorem 10.4 — would prohibit strong evaluation at $z = \omega^\delta = t^{-\delta}$, contradicting the hypothesis. $\square$

Remark 10.13 (This rigidity is a re-derivation, not a new theorem). Corollary 10.12 is presented as a consequence of the extension theorem, not as an independent contribution. The same conclusion is already in this collection twice over: the *Analysis* report proves all-scale polynomial rigidity directly, and the *Foundations* report records the exact boundary of the phenomenon with a pair of series over $\mathbb{C}((t^\mathbb{Q}))$ differing only in the sign of an exponent. Both source manuscripts merged here credit the repository’s pre-existing all-scale theme rather than claiming the corollary. The step that makes the reduction legitimate is the localization principle recorded in the foundations report — every *set* of surcomplex numbers lies in one divisible set-sized workspace — and the reduction consumes exactly that principle plus the cut δ . What is proposed as new here is only the exact cofinal/noncofinal criterion for a specified pair of set-sized workspaces (theorems 10.1 and 10.4) and the persistence of the Bézout obstruction under every cofinal extension (corollary 10.11). See section 13.

11 Assembly of the classifications

Proof of theorem 1.5. The ring is an integral domain because it is a subring of $K_\Gamma[[Z]]$. Its units are exactly the nonzero constants by corollary 6.3: a unit is zero-free, and nonzero constants are invertible. GCDs exist by corollary 7.5.

If Γ has no countable cofinal subset, corollary 3.3 gives $\mathcal{E}_\Gamma = K_\Gamma[Z]$, which is a PID over a field. If Γ has an order unit, its positive integer multiples form a countable cofinal subset, so nonpolynomial functions exist, and theorem 9.3 supplies the Bézout property. If Γ has countable cofinality and no order unit, theorem 8.4 supplies a proper ideal generated by two coprime functions. Such an ideal cannot be principal, since any principal generator would be a common divisor and hence a unit. Non-Noetherianity in both nonpolynomial cases is proposition 7.9.

These cases are mutually exclusive and exhaustive. For $\Gamma = 0$, strong summability at 1 says that only finitely many coefficient supports may contain 0, hence only finitely many coefficients are nonzero. $\square$

Proof of theorem 1.6. Part (a) is theorem 10.4 together with lemma 10.3; part (b) is corollary 10.5, whose coarse form is theorem 10.1; part (c) is theorem 10.7; part (d) is theorem 10.9; part (e) is corollary 10.10. The residue field statement is lemma 10.3. $\square$

Value-group condition	Entire series	Ideal structure
No countable cofinal subset	Only polynomials	PID
An order unit exists	Nonpolynomial series exist	Bézout, non-Noetherian
Countable cofinality, no order unit	Nonpolynomial series exist	GCD, not Bézout; two-generated proper ideal with no common zero

12 Explicit higher-rank and surreal examples

12.1 An explicit rank-infinite subgroup of the surreal field

Let

$$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}e_j, \quad e_j = \omega^j \in \mathbf{No}, \quad (28)$$

with the order inherited from $\mathbf{No}$: every element has finite support, and the largest index with a nonzero coefficient determines its sign. Thus $e_{j+1} \gg_A e_j$ and (e_j) is cofinal, so $\text{cf}(\Gamma_\infty) = \aleph_0$. No element is an order unit: the largest index in its finite support can always be exceeded. This is case (iii) of [theorem 1.5](#), and by [corollary 4.2](#) also $\mathcal{T}_{\Gamma_\infty}^\times = K_{\Gamma_\infty}^\times$.

The Bézout counterexample realized

Take $\gamma_n = e_n$ for $n \geq 1$, a positive cofinal sequence with $\gamma_{n+1} \gg_A \gamma_n$, as [lemma 8.1](#) demands. Using $t^\gamma = \omega^{-\gamma}$, [theorem 8.4](#) gives roots

$$\rho_n = \omega^{\omega^n}, \quad \sigma_n = \omega^{\omega^n} + \omega^{-\omega^{n+1}} \quad (n \geq 1), \quad (29)$$

and the functions

$$f(Z) = \prod_{n \geq 1} (1 - \omega^{-\omega^n} Z) = \prod_{n \geq 1} (1 - t^{e_n} Z), \quad (30)$$

$$g(Z) = \prod_{n \geq 1} \left(1 - \frac{Z}{\omega^{\omega^n} + \omega^{-\omega^{n+1}}} \right) \quad (31)$$

are entire on the fixed surcomplex Hahn field K_{Γ_∞} , not on all of $\mathbf{No}[i]$. They have real Hahn coefficients and all the displayed roots are real surreal numbers; by [corollary 7.8](#) this is visible from the conjugation-invariance of their divisors. The failure of a Bézout identity remains true even when complex Hahn coefficients are allowed in A, B , and consequently also when A, B are restricted to real Hahn coefficients.

The valuation formula [\(20\)](#) becomes

$$v(g(\rho_n)) = e_{n+1} + (2 - n)e_n + \sum_{1 \leq j < n} e_j,$$

whose first values are

$$\begin{aligned} \lambda_1 &= e_2 + e_1, & \lambda_2 &= e_3 + e_1, \\ \lambda_3 &= e_4 - e_3 + e_2 + e_1, & \lambda_4 &= e_5 - 2e_4 + e_3 + e_2 + e_1. \end{aligned}$$

The negative lower-level coefficients do not affect domination: the coefficient of the highest basis element e_{n+1} is 1.

For additional coefficient information, the k th coefficient of f is

$$[Z^k]f = (-1)^k \sum_{1 \leq j_1 < \dots < j_k} t^{e_{j_1} + \dots + e_{j_k}}, \quad v([Z^k]f) = e_1 + \dots + e_k,$$

the minimum coming from the unique subset $\{1, \dots, k\}$, whose coefficient is $(-1)^k$ and therefore cannot cancel. Division by the finite integer k still leaves a cofinal sequence of Archimedean classes, which independently exhibits the coefficient growth required by [theorem 3.1](#). The reciprocal $1/\sigma_j$ has the explicit support-certified form

$$\frac{1}{\sigma_j} = t^{e_j} \sum_{r \geq 0} (-1)^r t^{r(e_j + e_{j+1})},$$

a valid positive-support geometric identity by [lemma 2.1\(c\)](#). Its support need not itself be cofinal in the rank-infinite group; that is irrelevant to the legitimacy of this single coefficient. Entirety of the product is supplied by the cofinal escape of the factor indices j , not by a false claim about $r(e_j + e_{j+1})$.

The same workspace seen from its canonical product

The natural product over the *whole* basis,

$$P(Z) = \prod_{j \geq 0} (1 - t^{e_j} Z), \quad (32)$$

differs from [\(30\)](#) by the single linear factor of index zero:

$$P(Z) = (1 - t^{e_0} Z) f(Z) = (1 - \omega^{-1} Z) f(Z). \quad (33)$$

It is therefore an equally explicit member of the same ring, with one extra zero at ω .

Theorem 12.1 (A genuine infinite-rank surcomplex entire function). *The product [\(32\)](#) belongs to $\mathcal{E}_{\Gamma_\infty}$ and is nonpolynomial. Its zeros are exactly*

$$r_j = t^{-e_j} = \omega^{\omega^j} \quad (j \geq 0), \quad (34)$$

all simple and positive surreal. Its coefficient of Z^k is

$$c_k = (-1)^k \sum_{j_1 < \dots < j_k} t^{e_{j_1} + \dots + e_{j_k}}, \quad v(c_k) = e_0 + e_1 + \dots + e_{k-1} \quad (35)$$

for $k \geq 1$, with $c_0 = 1$. At a scale $s \in \Gamma_\infty$ put

$$N_s = \#\{j : e_j + s < 0\}, \quad M_s = \#\{j : e_j + s = 0\}.$$

These are finite, $M_s \in \{0, 1\}$, and

$$m_s(P) = \sum_{e_j + s < 0} (e_j + s), \quad (36)$$

$$I_s(P)(X) = (-1)^{N_s} X^{N_s} (1 - X)^{M_s}. \quad (37)$$

Finally

$$v(P'(r_j)) = e_j + \sum_{k < j} (e_k - e_j). \quad (38)$$

Proof. Since $e_j \rightarrow +\infty$ cofinally, the proposed zero divisor is radially finite, and [theorem 7.3](#) gives the product's existence, entireness and exact simple zero set. Expanding over finite subsets gives (35); among the k -element subsets, the sum of the first k basis elements is uniquely least in the largest-index order and its coefficient is $(-1)^k$, so it cannot cancel and every c_k is nonzero.

The slope sequence $(e_0 + \dots + e_{k-1})/k$ is cofinal: once $k-1$ exceeds the largest index occurring in a prescribed group element, its positive e_{k-1} coordinate dominates that element. This is a second, direct verification of entireness via [theorem 3.1](#).

At scale s only finitely many factors have $e_j + s \leq 0$. A factor of negative value contributes that value to m_s and contributes $-X$ to the normalized initial polynomial; a factor of value zero contributes $1 - X$; all positive factors contribute 1. This proves (36)–(37).

Differentiating the factor of index j and evaluating at r_j kills all other product-rule terms, giving

$$P'(r_j) = -t^{e_j} \prod_{k \neq j} (1 - t^{e_k - e_j}).$$

The factors with $k > j$ form a positive-support unit, and the finitely many factors with $k < j$ have valuations $e_k - e_j$; adding these proves (38). $\square$

This example is confined to no single Archimedean scale, yet its entire theory is nontrivial and its zero divisor completely explicit. Together with [remark 4.5](#) it shows that nonpolynomial entire functions and nonconstant restricted units are genuinely different existence questions.

12.2 Two series with the same leading valuations and different boundary behaviour

Let $\Gamma = \mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$ with $0 < \epsilon \ll \eta$ and define

$$A(Z) = \sum_{n \geq 1} t^{n^2\epsilon} Z^n, \quad (39)$$

$$B(Z) = \sum_{n \geq 1} (t^{n^2\epsilon} + t^{\eta - n^2\epsilon}) Z^n. \quad (40)$$

Their nonzero coefficients have exactly the same valuations $n^2\epsilon$. Nevertheless their *monomial* strong evaluation domains differ. For $s = a\eta + b\epsilon$ with $a, b \in \mathbb{Q}$,

	$a < 0$	$a = 0$	$a > 0$	
$A(t^s)$	not strong	strong	strong	
$B(t^s)$	not strong	not strong	strong	(41)

For $a > 0$ the leading valuations have η coordinate $na \rightarrow +\infty$, and [lemma 2.2](#) proves both positive assertions. For $a < 0$ those leading valuations strictly decrease. For $a = 0$ the exponents of $A(t^s)$ are $(n^2 + nb)\epsilon$, which after finitely many terms strictly increase with only finite repetition at any exponent, so they form a well-ordered support. The extra exponents of $B(t^s)$ are $\eta + (nb - n^2)\epsilon$, which strictly decrease for all sufficiently large n ; they destroy strong summability without affecting a single leading coefficient valuation.

Neither series is entire over the whole rank-two field: their coefficient slopes $n\epsilon$ remain below η , so [theorem 3.1](#) correctly excludes both. There is no tension between this example and the valuation-only criterion. Leading valuations can fail to decide strong summability at a *boundary* scale, while global summability at *every* scale removes the ambiguity. Formula (41) is stated for monomial arguments only; it does not silently assert a classification for arbitrary nonmonomial arguments of B .

12.3 One extension preserves the function; another destroys entireness

Adjoin $\omega^{1/2}$ to Γ_∞ and take the divisible additive span. This is an ordered-group extension in which Γ_∞ remains cofinal, since a sufficiently large integer power of ω dominates every finite combination of the added and old generators. The series P , and likewise the pair f, g , remain entire by [corollaries 10.5 and 10.11](#); in particular the Bézout failure persists.

Instead adjoin the new generator $e_* = \omega^\omega > \Gamma_\infty$ and set $\Delta = \Gamma_\infty \oplus \mathbb{Q}e_*$. At the point $Z = t^{-e_*}$ the nonzero summands of P have leading exponents $(e_0 + \cdots + e_{k-1}) - ke_*$, whose consecutive differences are $e_k - e_* < 0$; the defining family is therefore not strongly summable, exactly as [theorem 10.1](#) predicts. [Theorem 10.9](#) rules out any alternative entire power series over K_Δ agreeing with P on the old field, or even on $\mathbb{C}$. On the still-defined domain $\mathcal{D}_{\Gamma_\infty, \Delta}$, however, P is unchanged and its zeros remain exactly (34), by [theorems 10.4 and 10.7](#). This concrete pair is the clearest illustration of [remark 10.6](#).

12.4 Why finite rank does not give the counterexample

For $\Gamma = \mathbb{Q}^d$ in lexicographic order let h lie in the largest Archimedean class. Integer multiples of h are cofinal, so h is an order unit and the entire ring is Bézout by [theorem 9.3](#), even when $d > 1$. The same holds for $\mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $e_2 \gg_A e_1$: take $h = e_2$. It is *absence of a greatest Archimedean class*, together with countable cofinality, that forces the negative case — not high rank.

12.5 An uncountable hierarchy gives polynomial rigidity

Let

$$\Gamma_{\omega_1} = \bigoplus_{\alpha < \omega_1} \mathbb{Q}e_\alpha, \quad e_\alpha = \omega^\alpha, \quad (42)$$

again ordered by the largest nonzero index. Its cofinality is $\aleph_1$: the basis is cofinal, whereas any countable family of finite supports has its indices bounded below ω_1 and a larger basis element bounds the whole family. Therefore

$$\mathcal{E}_{\Gamma_{\omega_1}} = \mathcal{T}_{\Gamma_{\omega_1}} = K_{\Gamma_{\omega_1}}[Z], \quad (43)$$

the middle equality because a nonpolynomial restricted series would itself supply a countable sequence of coefficient valuations tending cofinally to infinity. This example shows that infinite rank by itself does not even guarantee nonpolynomial entire functions: rank is not the driver, the order type above the scales is.

12.6 The separation table

Value group	cf	Has an order unit	Nonpolynomial entire series	Nonconstant restricted units
$\mathbb{Q}$ or $\mathbb{R}$	$\aleph_0$	Every positive element	Yes	Yes
$\mathbb{Q}\epsilon \oplus \mathbb{Q}\eta$, $\epsilon \ll \eta$	$\aleph_0$	Yes; not every positive element	Yes	Yes, with the sharper gap test
Γ_∞	$\aleph_0$	No	Yes	No
Γ_{ω_1}	$\aleph_1$	No	No	No

The second and third rows are the two nonpolynomial cases of [theorem 1.5](#); the third is the non-Bézout one. Note that $\mathbb{R}$ is uncountable of rank one with $\text{cf} = \aleph_0$, while Γ_∞ has infinite rank with $\text{cf} = \aleph_0$: neither cardinality nor rank is the invariant at work.

13 Relation to other reports in this collection

This report is a merge of two independently written manuscripts. Both were prepared against pinned snapshots of the same repository, and both state explicitly which of their neighbours' results they do *not* claim. Those statements are preserved here, and the cross-references are reciprocal.

Rank-one Berkovich geometry — this report extends it. The report *From Surcomplex Hahn Fields to Berkovich Geometry* [14] fixes $K = \mathbb{C}((t^{\mathbb{R}}))$: the value group is $\mathbb{R}$, the field is spherically complete, and $|a|_v = \exp(-v(a))$ is a genuine real-valued non-Archimedean absolute value. It defines intrinsic convergence using that real-valued absolute value. Its entire-function section proves zero-free entire rigidity and a convergent canonical product *there*, and neither source manuscript merged here claims to originate either point. The present report is the arbitrary-rank generalization. Three specific relations should be recorded.

- *Where rank one sits in the classification.* For $\Gamma = \mathbb{R}$ we have $\text{cf}(\mathbb{R}) = \aleph_0$ and 1 is an order unit, so rank one falls squarely in case (ii) of [theorem 1.5](#): nonpolynomial entire functions exist, $\mathcal{E}_{\mathbb{R}}$ is a non-Noetherian Bézout domain, and arbitrary radially discrete Hermite interpolation is available. The pathology of case (iii) is invisible at rank one.
- *Why rank one is not general.* That report's `prop:rankobstruction` proves that $\mathbb{Q} + \mathbb{Q}\omega$ admits no order-preserving *injective* additive map to $\mathbb{R}$. That is exactly why [lemma 9.1](#) constructs a *coarsening* with a possibly nonzero kernel H rather than an embedding, and why it is available only when an order unit exists. The two statements are complementary: one forbids faithful scalarization in general, the other supplies the best available non-faithful substitute in the order-unit case.
- *The ring chains resemble each other and are not equal.* That report's strict chain $K[Z] \subsetneq \mathcal{T} \subsetneq \mathcal{P} \subsetneq \mathcal{A} \subsetneq \mathcal{F} \subsetneq \mathcal{E}^-$ locates its polynomial-coefficient and formal algebras. Our chain (4) resembles it but is not the same chain, and the symbols $\mathcal{T}$, $\mathcal{A}$, $\mathcal{E}$ do not denote the same rings in the two reports. [Convention 1.2](#) forbids transferring a statement across that boundary.

Conversely, the rank-one report's own entire-function section should be read with the knowledge that the arbitrary-rank classification now exists and that its rank-one instance lands in the Bézout case.

Analysis and Foundations — this report is published beside them. [Corollary 10.12](#) re-derives a conclusion the collection already contains twice: the *Analysis* report proves all-scale polynomial rigidity directly (its Theorem `f:thm-rigidity`), and the *Foundations* report records the exact boundary of the phenomenon with a pair of series over $\mathbb{C}((t^{\mathbb{Q}}))$ that differ only in the sign of an exponent, noting that $\sum t^{+n^2} z^n$ is strongly summable at every point of that field and is *nonpolynomial* while remaining coherent at every radius, and that this does not refute the all-scale theorem but shows that the all-scale theorem does not transfer to a *fixed* value group. That example is precisely [remark 3.2](#) of this report, and the trichotomy of [theorem 1.5](#) explains why it works over

$\mathbb{Q}$ — $\text{cf}(\mathbb{Q}) = \aleph_0$ and 1 is an order unit, so $\mathbb{Q}$ is a case (ii) group — and what it would take to break it, namely replacing $\mathbb{Q}$ by a group of uncountable cofinality. Conversely, our reduction in [corollary 10.12](#) consumes the foundations report’s *localization principle*: every set of surcomplex numbers lies in one divisible set-sized workspace. That principle is what makes the reduction legitimate, and it is cited here, not reproved.

Global divisors and analytic geometry — different rings. The global-divisor report [15] works in $\mathcal{O}(U)((t^\Gamma))$, with ordinary holomorphic coefficient functions on a common complex domain; its divisor, interpolation and Picard obstructions live in a different function ring from the strongly entire ring studied here, and nothing in this report contradicts or restates them. The analytic-geometry report states the ring discipline that [convention 1.2](#) invokes by name, for its own four coefficient rings. Neither the objects nor the theorems transfer.

14 What the results do and do not establish

14.1 Three independent distinctions

The first distinction is between strong Hahn summation and intrinsic valuation convergence. Positive well-ordered support suffices for the former; cofinal progress through every group threshold is needed for the latter. On a single disk they differ. For a series strongly evaluable at *every* point, [theorem 3.1](#) shows the distinction disappears within its original field.

The second is between countable cofinality and an order unit. Countable cofinality supplies a sequence of ever larger scales, possibly moving through infinitely many Archimedean classes; an order unit supplies all those bounds by the integer multiples of one element. Entire functions need only the first; a nonconstant restricted inverse requires the second. The group Γ_∞ separates them, and that separation is the whole content of the gap between cases (ii) and (iii) of [theorem 1.5](#).

The third is between an intrinsic topology in one workspace and the fine topology tested against every surreal scale. The same countable sequence may converge intrinsically in K_Γ and fail to converge after a noncofinal enlargement. None of our proofs identifies these topologies. See [section 2.3](#).

14.2 Established, and no more

The results isolate three logically different phenomena. Coefficient slopes determine whether an ordinary countable power series can be evaluated at every scale in a fixed field. Support-controlled preparation and canonical products completely describe zeros and divisibility within that category. The order structure above those scales determines whether arbitrary values and jets can be prescribed and whether coprime functions generate the unit ideal.

In particular the factorization monoid alone cannot decide the Bézout property. In both nonpolynomial cases a function up to a scalar is exactly a radially finite zero divisor, and every effective subdivisor is again principal; nevertheless only one of those cases admits unrestricted discrete interpolation. The additive structure of the ring, not merely its multiplicative divisibility relation, is essential.

The non-Bézout example is not a delicate choice of a transcendental ordinary complex function or of an uncomputable coefficient. Its nodes, displacements and factor coefficients are explicit Hahn

expressions over rational data. What cannot be replaced by a finite calculation is the quantification over all coefficient indices and all Archimedean scales.

14.3 Scope that must not be silently enlarged

The following restrictions are stated by the source manuscripts and are retained without weakening.

Hypotheses that cannot be relaxed. All supports and index sets are sets; there is no proper-class summation. The field is a *full* Hahn field with complex residue coefficients, and Γ is divisible. Algebraic closedness is used essentially, to split the prepared finite polynomial; it is the sole import of [theorem 10.7](#). A nondivisible group, or a support-restricted Hahn subfield, may lose the roots needed to split the prepared polynomial, so the factorization theorem cannot simply be copied there. The Hahn–Neumann support lemmas ([lemma 2.1](#)) and algebraic closedness [4] are imported inputs, not proved here.

Categories excluded. The ring is one-variable and strong-summation entire. It is *not* the ring of ordinary-holomorphic-coefficient Hahn families on one common complex domain, not a ring of arbitrary fine-local surcomplex germs, and not a several-variable analytic algebra. The arbitrary-rank preparation theorem is proved in $\mathbb{C}[X]((t^\Gamma))$, not asserted in a larger power-series coefficient ring. There is no contour integral here, no global surcomplex exponential at unrestricted imaginary arguments, and nothing about the Berarducci–Mantova derivation; the statement about primitives in [proposition 3.5](#) concerns termwise Z -integration with coefficients fixed and says nothing about differential equations in the coefficient field. All entire functions here are represented by one ordinary formal power series at the origin: translation preserves this class, but does not identify it with every locally analytic or surreal-differentiable function. Piecewise functions on fine-topological domains, Gonshor’s exponential, locally represented classes and other support or summation conventions lie outside the hypotheses, so [corollary 10.12](#) is *not* a general no-go theorem for surreal analysis.

Limits of the coarsening and of the extension theorem. The rank-one coarsening of [lemma 9.1](#) changes residue information even though it preserves the topology and the entire ring; residue-direction data is not identified across the two valuations. The scalar-extension results keep the coefficient field $\mathbb{C}$ fixed and use canonical ordered-group inclusions of full Hahn fields only: they are not theorems about arbitrary extensions of incomplete subfields, about changing residue fields, or about analytic sheaves. The hidden-tail table (41) is stated for monomial arguments and does not classify arbitrary nonmonomial arguments of B . The single-variable uniform-domain statement is *false* in several variables — a series independent of one coordinate imposes no constraint in that coordinate — and the correct multivariable replacement is posed below as an open question. No density of $\mathbb{C}$ in any fine surreal topology is asserted, and strong Hahn summability, intrinsic valuation convergence and the fine surreal topology are deliberately never identified.

What is and is not claimed about priority and verification. The results of this report are self-declared proposed original contributions, not a solution of any named published conjecture. Both source manuscripts state that priority is *not* certified and that the literature checks were targeted rather than exhaustive; in one of them several general web searches returned irrelevant results and were explicitly not treated as evidence of absence, and in the other a connector keyword search returning nothing was likewise not treated as proof. The rank-one factorization specialization

is stated to be classical; standard Hahn algebra, algebraic closedness and support lemmas are not claimed; Cherry himself calls his factorization results well known; the rank-two distinction between positive valuation and topological nilpotence is established [8, Section 6.2]; and the all-surcomplex polynomial corollary is credited to the repository, as recorded in [remark 10.13](#). One source’s audit further notes that Lazard was checked bibliographically only and that Conway and Gonshor were not re-audited in full; its repository review was a targeted read of four files, not a line-by-line proof audit or an independent build. Neither manuscript was independently refereed and neither was machine-checked: one records `formal_verification false`, `independent_peer_review false` and `priority_certified false` in its build record, and the other states flatly that no Lean verification is claimed. No theorem below depends for its validity on the correctness of an unrefereed repository report. The repository’s Lean modules on finite algebra and Hahn supports are plausible *prerequisites* for a future formalization, not a certificate that these proofs have been formalized; missing implementation is not treated here as an open mathematical problem. No repository files under `code/` or `data/` have been modified.

What the finite checks do not do. See [appendix C](#) for the full statement. In summary: the two delivered suites verify 831 and 664 finite exact assertions respectively, and they prove *no* infinite theorem. Both articles and both READMEs state that the finite truncations of the counterexample pair *are* coprime polynomials and *do* satisfy polynomial Bézout identities, so no finite computation can witness [theorem 8.4](#). The preparation checks run modulo t^9 in an integer exponent subgroup, not an arbitrary-rank group, and the finite dominance samples test only $k \in \{1, 2, 10, 100, 10000\}$, not all integer multiples. The checks cannot verify arbitrary well-ordered supports, the cofinality dichotomy, the necessity direction of [theorem 4.1](#), or any infinite product identity.

14.4 Questions opened by the classification

The following are research questions generated by this work, not claims that any particular publication lists them as open.

Question 14.1 (Exact interpolation image). When Γ has countable cofinality but no order unit, characterize the image of

$$\mathcal{E}_\Gamma \longrightarrow \prod_{x \in \text{supp } D} K_\Gamma[U]/(U^{D(x)})$$

for a prescribed radially finite divisor D .

This question is open on the article’s own terms. The growth barrier gives explicit necessary inequalities and proves that the image can be proper, but sufficiency for arbitrary nodes and jets is not established. A satisfactory answer should encode the compatibility of divided differences across convex subgroups, rather than only the size of individual values.

Question 14.2 (Finitely generated ideals beyond their common divisor). Can one describe the ideal (f, g) by its common zero divisor together with an explicit obstruction in an interpolation quotient?

There is already an exact reduction: after dividing out the GCD, the Bézout problem is whether the inverse jets of one function along the other’s zero divisor lie in the interpolation image. What is missing is a usable intrinsic description of that image. [Theorem 8.4](#) proves that a zero-divisor description alone is insufficient.

Question 14.3 (Several variables). Which parts of the cofinality criterion and of the Bézout obstruction extend to several ordinary variables, and which local preparation statements retain explicit Hahn support certificates? In particular, since the single-variable uniform extension domain is false in several variables, what is the correct replacement — presumably one depending on the unbounded degree directions of the support, including the boundary where higher Hahn tails matter?

One must separate the cofinal growth of total degree from the geometry of positive-dimensional zero sets. Cherry’s several-variable work is an important rank-one antecedent [6]; no arbitrary-rank multivariable GCD theorem is established here. [Appendix B](#) proves only the finite-variable convergence and unit criteria, and it does not extend the one-variable divisor classification to higher-dimensional zero sets.

Question 14.4 (Restricted and incomplete subfields). How much of the divisor theory survives for incomplete surreal subfields or for support-restricted Hahn subfields? Our proofs isolate two requirements: closure under the positive-support preparation construction, and enough algebraic roots to split the finite prepared polynomials. A classification of natural surreal subfields meeting both would turn these full-Hahn results into a more computationally selective theory.

14.5 A proof-assistant route

The research statement does not require a formalization of the entire proper class **No**. A first formal development can work over a set-sized divisible linearly ordered abelian group and a Hahn field. The most useful first target is the global criterion [theorem 3.1](#), whose nontrivial implication is a short ordered-group argument plus the definition of strong summability; the next is the coarsened polynomial argument for [theorem 4.1](#). Support preparation should be formalized only after the finite-word Neumann lemma and arbitrary regrouping are available. The explicit obstruction then requires only ordered-group inequalities, finite product valuations and evaluation homomorphisms. This is a dependency plan, not a formal-verification claim. The current repository ledger `docs/FORMALIZATION.md` distinguishes checked Hahn support, summability and algebraic-closure prerequisites from the still-pending statements of this report. No Lean code is included in this report package.

15 Conclusion

The threshold for nonpolynomial entire behaviour in a Hahn–surcomplex workspace is countable cofinality, not rank one. Among workspaces admitting nonpolynomial entire series, the universal Bézout property and full radially finite Hermite interpolation require the stronger condition that an order unit exist. This is also the threshold for existence of a nonconstant restricted unit; for a particular restricted series, its least nonconstant coefficient gap relative to the constant term must itself be a positive order unit. Individual coprime pairs may admit a Bézout identity even when the universal property fails. The two existence conditions separate in an explicit infinite-rank subgroup of the surreal numbers, and the separation is witnessed by two entire functions with disjoint simple zero sets whose ideal is proper.

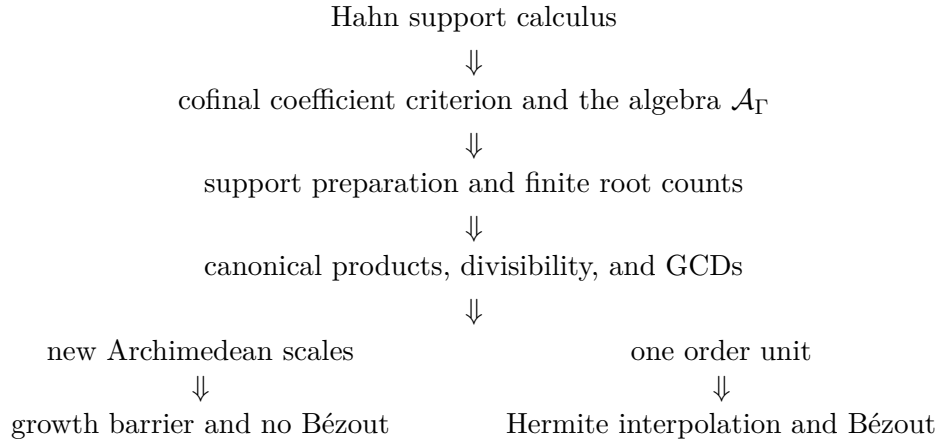
After enlarging the value group, every nonpolynomial entire series has the same exact strong domain: the valuation ring of the convex hull of its original scales. Cofinality is precisely what preserves entireness, while support preparation preserves every existing zero and prevents new ones throughout the surviving domain. The canonical product theorem then shows that coefficients,

zeros and the extension boundary encode the same cofinal scale structure. Together this gives one explanation of both nonpolynomial fixed-workspace functions and polynomial rigidity on the whole surcomplex class, without confusing Hahn identities with convergence in the fine surreal topology.

A Proof-dependency and hypothesis audit

A.1 The dependency chain

The main proof can be read along the following chain.



The arrow from preparation to root counting additionally uses algebraic closedness of the Hahn field. The arrow to interpolation uses completeness of the real-valued coarsening, proved directly in [lemma 9.1](#). No proof of the negative result depends on the positive interpolation theorem.

The scalar-extension results of [section 10](#) sit on a deliberately separate branch. The restricted-unit theorem ([theorem 4.1](#)) uses Gauss multiplicativity (5) and a convex coarsening only; it uses neither zero counting nor canonical products. The universal-domain theorem ([theorem 10.4](#)) uses the global criterion and integral evaluation; it does not use factorization. *Only* zero conservation ([theorem 10.7](#)) imports both algebraic closedness and preparation. The divisor classification is downstream of the local tools and the elementary entire algebra. Thus the principal classifications are not consequences of an unproved global factorization assertion, and conversely the factorization proof never assumes that an arbitrary zero-free restricted series is restricted-invertible — by [example 4.3](#) that is exactly the assertion that can fail in higher rank.

A.2 Where an attained minimum is load-bearing

Attained minima. The proof of [theorem 3.1](#) uses a minimum attained at an ordinary *finite* coefficient index, not a termwise inequality. The proof of [lemma 8.2](#) uses an attained minimum in the same way: termwise strict inequalities alone would not justify a strict inequality for an arbitrary limiting process. These are the two places in the article where the distinction is load-bearing, and neither can be replaced by an estimate valid only term by term.

Two independent infinities. The finite index k in a power series is not a Hahn exponent. Integer multiples of a positive exponent need not be cofinal. The support lemma [lemma 2.1\(b\)](#) handles geometric-series expansions without that property; the positive interpolation proof uses it only

after constructing a legitimate real-valued coarsening, which is available only under the order-unit hypothesis.

Divisibility is not assumed in its own proof. The quotient F/G in [theorem 7.4](#) is first formed, after removing the shared zero at the origin, in $K_\Gamma[[Z]]$. Only preparation at every scale proves it entire. This avoids assuming the global divisibility theorem in its own proof.

The counterexample is irreducibly infinite. No finite truncation of the pair (19) can witness non-Bézout behaviour: the finite truncations are polynomials with disjoint root sets and therefore generate the unit ideal. The contradiction requires a single candidate coefficient lower bound to fail at cofinally many successively dominating scales.

Why preparation must be a support recursion. [Corollary 5.3](#) — the fact that the prepared factorization is unchanged by enlarging the value group — is available precisely because preparation was proved by well-founded recursion over S^* . An iteration whose convergence had been asserted in K_Γ would have to have that convergence re-established in K_Δ , and [example 4.3](#) shows it can genuinely fail.

B Finite-variable convergence and unit criteria

Let $d \geq 1$ be finite, write $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ and $|\alpha| = \alpha_1 + \dots + \alpha_d$. Define

$$\mathcal{T}_{\Gamma,d} = \left\{ \sum_{\alpha} a_{\alpha} X^{\alpha} : v(a_{\alpha}) \rightarrow +\infty \text{ as } |\alpha| \rightarrow \infty \right\},$$

and define $\mathcal{E}_{\Gamma,d}$ by strong summability of the *whole* family $(a_{\alpha} x^{\alpha})_{\alpha \in \mathbb{N}^d}$ for every $x \in K_{\Gamma}^d$. Summing first along one coordinate and then another is not the definition.

Theorem B.1. *For a finite-variable formal series $F = \sum_{\alpha} a_{\alpha} X^{\alpha}$,*

$$F \in \mathcal{E}_{\Gamma,d} \iff \frac{v(a_{\alpha})}{|\alpha|} \rightarrow +\infty \quad (|\alpha| \rightarrow \infty, |\alpha| > 0).$$

It suffices to test strong summability at the diagonal monomial points $(t^s, \dots, t^s)$ for all $s \in \Gamma$. Nonpolynomial members exist exactly when $\text{cf}(\Gamma) = \aleph_0$.

Proof. Assume strong summability at every diagonal monomial point and $F \neq 0$. At $(1, \dots, 1)$ it gives the attained lower bound $m = \min_{a_{\alpha} \neq 0} v(a_{\alpha})$. If the slope condition fails, some $h \in \Gamma$ has violating indices of arbitrarily large total degree $N = |\alpha|$, with $a_{\alpha} \neq 0$ and $v(a_{\alpha}) \leq Nh$. Choose $q > \max\{0, |m|, |h|\}$. Strong summability at $(t^{-2q}, \dots, t^{-2q})$ makes the set of leading exponents $v(a_{\alpha}) - 2|\alpha|q$ well ordered. Its minimum μ is attained at an index β , and

$$\mu = v(a_{\beta}) - 2|\beta|q \geq m - 2|\beta|q > -(2|\beta| + 1)q.$$

A violating index of degree $N > 2|\beta| + 1$ instead gives

$$v(a_{\alpha}) - 2Nq \leq Nh - 2Nq < -Nq < \mu,$$

a contradiction. This is the finite-index minimum argument of [theorem 3.1](#); it does not require multiples of q to be cofinal. The zero series satisfies the criterion directly.

Conversely, for a point with all coordinates nonzero set $s = \min_i v(x_i)$; then $v(a_\alpha x^\alpha) \geq v(a_\alpha) + |\alpha|s \rightarrow +\infty$, and because bounded total degrees contain finitely many indices, [lemma 2.2](#) applies after any enumeration respecting total degree. If some coordinates vanish, discard the vanishing terms and apply the same argument to the remaining coordinates. The cofinality assertion follows exactly as in one variable; for existence use an entire nonpolynomial in X_1 alone. $\square$

Theorem B.2. *A nonconstant $F \in \mathcal{T}_{\Gamma,d}$ is a unit of that algebra if and only if $a_0 \neq 0$ and $\min_{\alpha \neq 0, a_\alpha \neq 0} v(a_\alpha/a_0)$ is a positive order unit of Γ .*

Proof. Regrouping into $\mathbb{C}[X_1, \dots, X_d]((t^\Gamma))$ gives multiplicativity of the Gauss valuation. If $FG = 1$, normalize both constant coefficients to 1; their Gauss values are nonpositive and sum to zero, so their initial polynomials multiply to 1 and are constant, and all nonconstant coefficient values are positive. If their minimum δ is not an order unit, coarsen by the proper convex subgroup (8) it generates: restrictedness reduces both series to finite polynomials over the coarsened residue field, and at least one nonconstant coefficient of F survives, contradicting invertibility of its reduction. Conversely an order-unit gap makes the geometric inverse restricted, with precisely the tail estimates of [theorem 4.1](#); finite-variable convolution has finitely many terms at each multi-index, so those estimates are unchanged. $\square$

Finite-dimensionality is used in the finiteness of bounded-degree index sets. These statements assert nothing for infinitely many variables, and they do not extend the one-variable divisor classification to higher-dimensional zero sets. As noted in [section 14.3](#), the single-variable uniform extension domain of [theorem 10.4](#) is *false* in several variables.

C Reproducible finite checks

Two independent exact-arithmetic suites accompany this report, preserved unmodified in `code/` and `data/`. Both use only the Python standard library, with exact `fractions.Fraction` arithmetic and no floating point, and both represent group elements as tuples of rational coordinates ordered by the largest nonzero coordinate.

Suite A (831 checks). The source program `verify_examples.py` is shipped as

`code/06-cofinality-bezout-trichotomy-verify-examples.py`

It checks finite instances of the reverse-lexicographic exponent inequalities, the valuation formula (20) in the first nine coordinates of Γ_∞ , leading exponents of finite canonical products, a finite-truncation implementation of the preparation recursion (10)–(11), and the local inverse-jet identity underlying (22). The delivered run passed all **831** checks, with category counts 72 paired-root checks, 40 finite dominance samples, 88 product-coefficient checks, 456 preparation checks and 175 inverse-jet checks; the record is the delivered verification JSON. The script takes a *required* output path argument and exits nonzero without one; a fresh output path preserves the historical record. A review run on a temporary copy passed 831 of 831 again.

Suite B (664 checks). The source program `verify.py` is shipped as

`code/03-cofinality-and-scalar-extension-verify.py`

It checks finite truncations of (32) for $N = 1, \dots, 8$ — each coefficient’s leading exponent, its binomial subset count and its signs — exactness and simplicity of each root t^{-e_j} and the simple-root derivative (38), closed- and open-ball Newton counts at sampled scales including half-shifts by e_0 , descent of the valuations after adjoining the dominant scale e_9 , the rank-two hidden-tail comparison of section 12.2 for $n = 1, \dots, 100$ and $\lambda \in \{-7, -\frac{1}{2}, 0, \frac{3}{2}, 10\}$, and the preparation recursion for (12) through perturbation order eight. The delivered run reports **664** passing checks. A review run on a temporary copy also passed all 664 checks, leaving the historical records unchanged.

What the checks cannot do. Both suites print their own scope disclaimers, and both are correct to do so. The counts are counts of finite assertions, not confidence scores for any infinite theorem. Neither suite can prove cofinality statements, Hahn support well-ordering for every permitted group, any infinite product identity, the necessity direction of theorem 4.1, or the nonexistence of a global Bézout solution. In the preparation checks the t -adic truncation is explicitly finite, so higher coefficients are discarded only after the claimed congruence has been specified, and the exponent subgroup used is an integer one rather than an arbitrary-rank group. The finite dominance samples test only $k \in \{1, 2, 10, 100, 10000\}$. Most importantly, the finite truncations of (19) *are* coprime polynomials and *do* satisfy polynomial Bézout identities, so no finite computation can witness theorem 8.4. What the checks can detect is an indexing error, a sign error in a paired root valuation, a reversed order convention, a wrong root-count endpoint or a wrong finite preparation coefficient. Everything else rests on the proofs in the body.

D Source and novelty ledger

This report merges two manuscripts, both dated 21 September 2026, each of which pinned the repository at its own snapshot: one at commit `4896a2808ce30e01b1c86ae3ba2295a64762d246` and the other at `aa846271b4dcae2c055b216126a87210292ec19b`. The first re-read four files at its pinned commit — the root README, the documentation catalogue, and the READMEs of the rank-one and global-divisor reports — after its working branch changed during preparation; the second cited the documentation catalogue and the rank-one article. Both reviews were targeted reads, not line-by-line proof audits or independent builds. Neither reinterprets missing Lean implementation as an unsolved mathematical problem.

The primary literature checked across the two manuscripts comprises Poonen’s Hahn-field construction and algebraic-closedness corollary, Neumann’s ordered division-ring paper for the support calculus, Cherry’s complete real-valued non-Archimedean setting together with his one-variable factorization statement and his lectures, Conrad’s higher-rank valuation examples, Lazard’s bibliographic record as the historical reference for one-variable zeros, van der Hoeven’s generalized-series operator framework, and the Conway–Gonshor surreal normal-form background with cross-checks against [10, 11]. Lazard was checked bibliographically only, and Conway and Gonshor were not re-audited in full. General web searches were attempted; several returned irrelevant results and were not treated as evidence that a theorem is absent from the literature.

The strongest proposed original statements are theorem 8.4 and its use in theorem 1.5, together with theorems 10.4, 10.7 and 10.9 and their assembly in theorem 1.6. The sharp workspace criterion and the persistence result are theorem 10.1 and corollary 10.11; the restricted-unit criterion

[theorem 4.1](#) is closely related to established topological nilpotence distinctions, and it is its precise necessity-and-sufficiency formulation for [\(3\)](#), including groups with no order unit, that is proposed here. The rank-one factorization formula is explicitly classical. The all-surcomplex polynomial corollary is credited to the pre-existing all-scale theme in the repository ([remark 10.13](#)). A later discovery of an antecedent would change the priority discussion, not the content or the hypotheses of the displayed proofs.

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- [14] *From Surcomplex Hahn Fields to Berkovich Geometry: rank-one convergence, Tate algebras, analytic annuli, and an entire-function case study*, research report in the *Surreal* repository, docs/surcomplex/rank-one-berkovich/, inspected 21 September 2026. Already contains rank-one zero-free rigidity and a canonical product theorem; its prop:rankobstruction is the rank-one boundary discussed in section 13.

- [15] *Global Support Obstructions: Moving Divisors, Mittag-Leffler, and a Picard Dichotomy*, research report in the *Surreal* repository, [docs/surcomplex/global-divisors/](#), inspected 21 September 2026. Stated over $\mathcal{O}(U)((t^\Gamma))$, a different coefficient ring.
- [16] *Analytic Geometry over Surcomplex Hahn Fields*, research report in the *Surreal* repository, [docs/surcomplex/analytic-geometry/](#). Source of the ring-discipline convention invoked in [convention 1.2](#).
- [17] *Surcomplex Analysis*, research report in the *Surreal* repository, [docs/surcomplex/analysis/](#). Contains the all-scale polynomial rigidity theorem discussed in [remark 10.13](#).
- [18] *Foundations*, research report in the *Surreal* repository, [docs/foundations-and-computation/foundations/](#). Source of the localization principle and of the sign-flipped pair $\sum t^{\pm n^2} z^n$ discussed in [remark 3.2](#) and [section 13](#).